Deep Learning for Computer Vision

Lecture 1: Introduction

Deep Learning for Computer Vision

Deep Learning for Computer Vision

Building artificial systems
that process, perceive, and
reason about visual data

Computer Vision is everywhere!













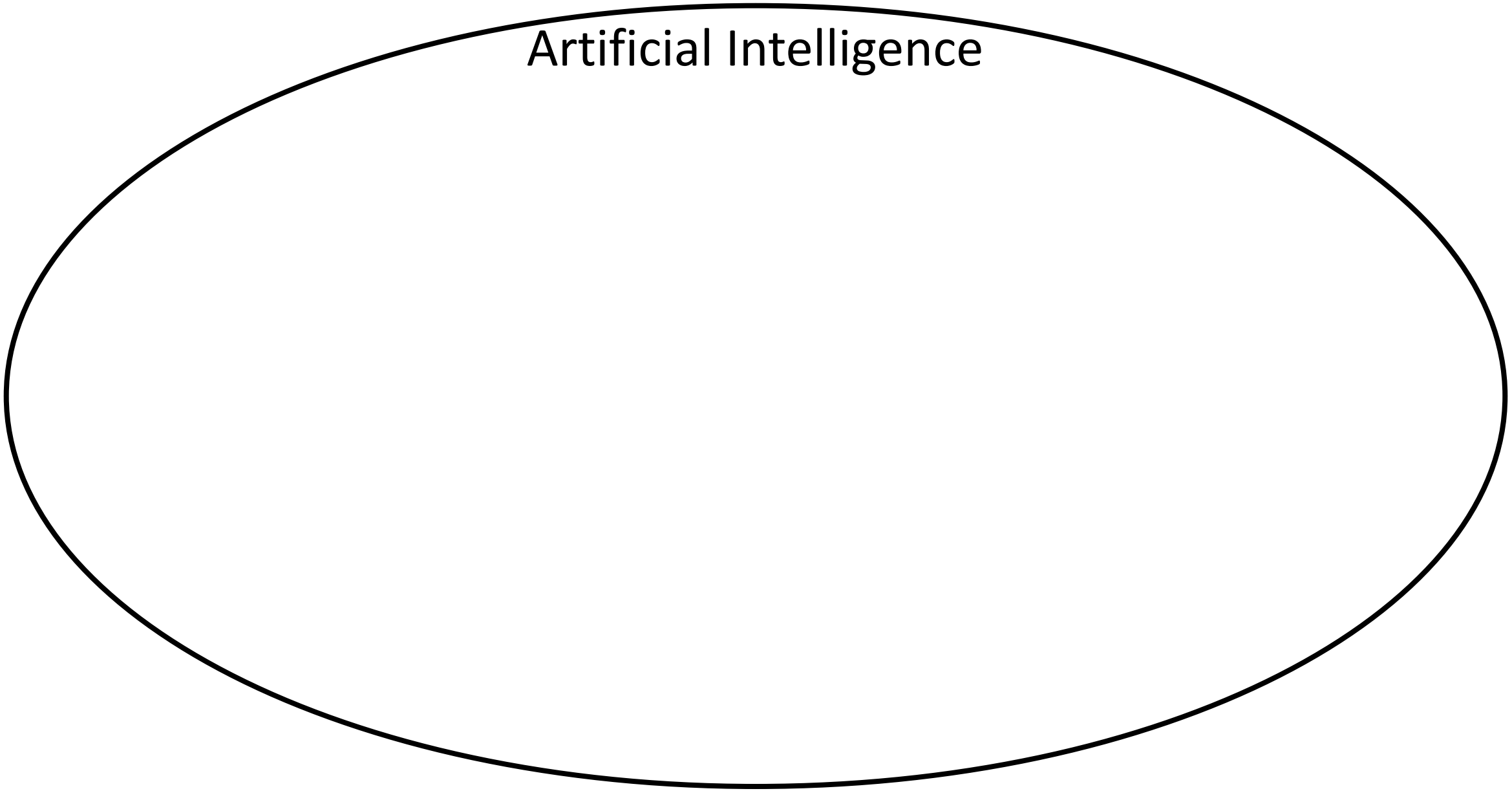


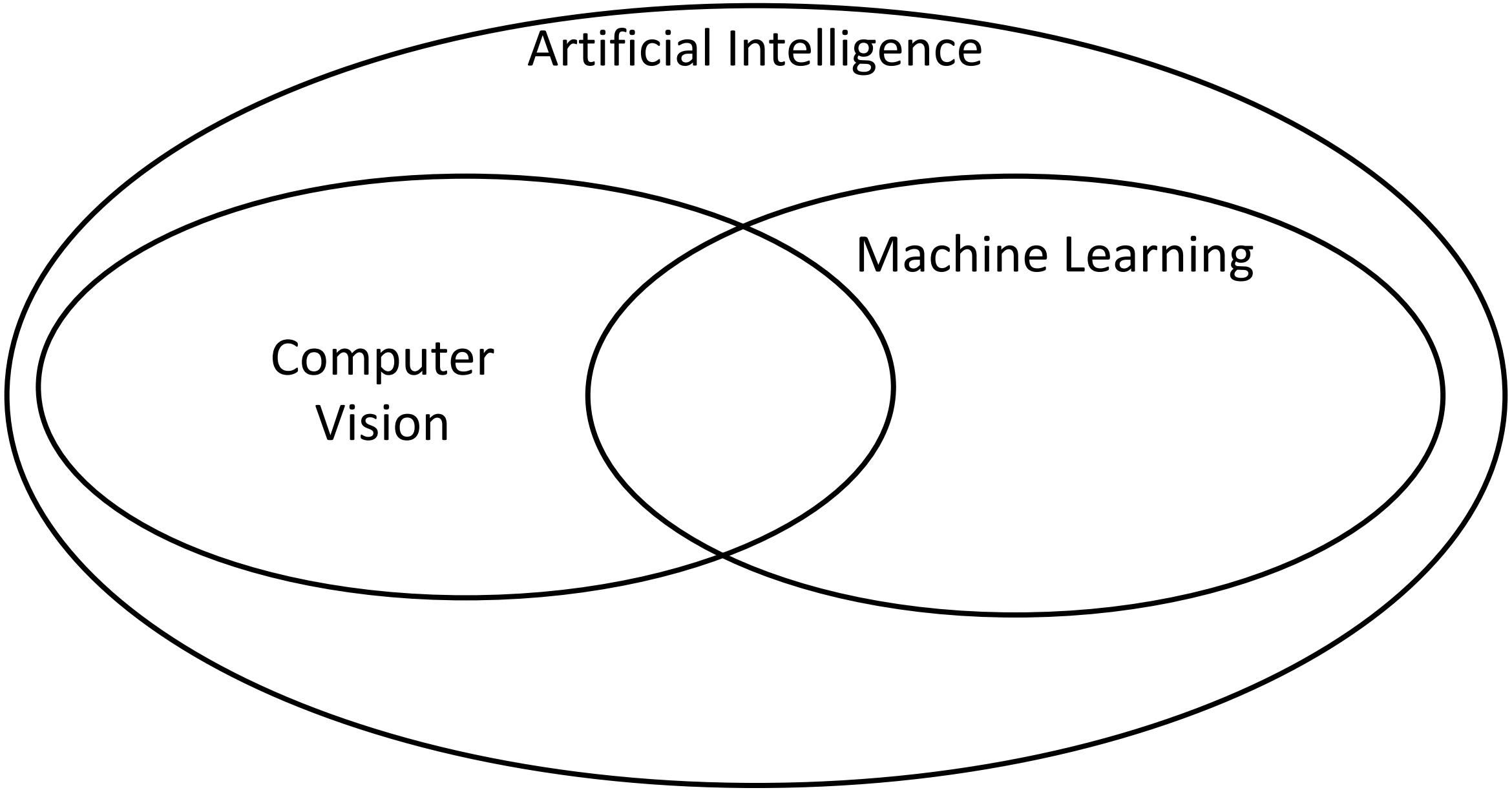

Deep Learning For Computer Vision

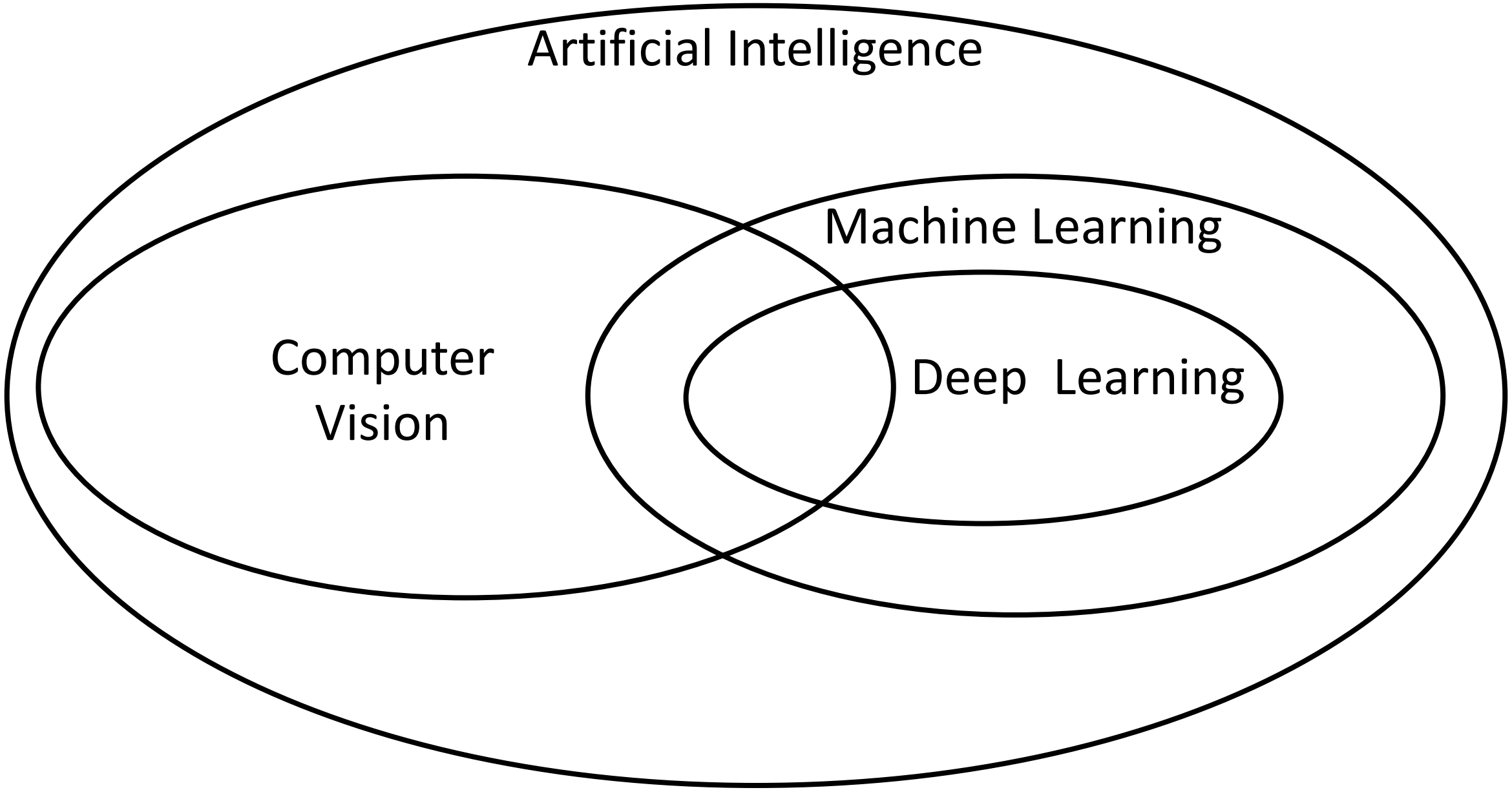
Building artificial systems that
learn from data and experience

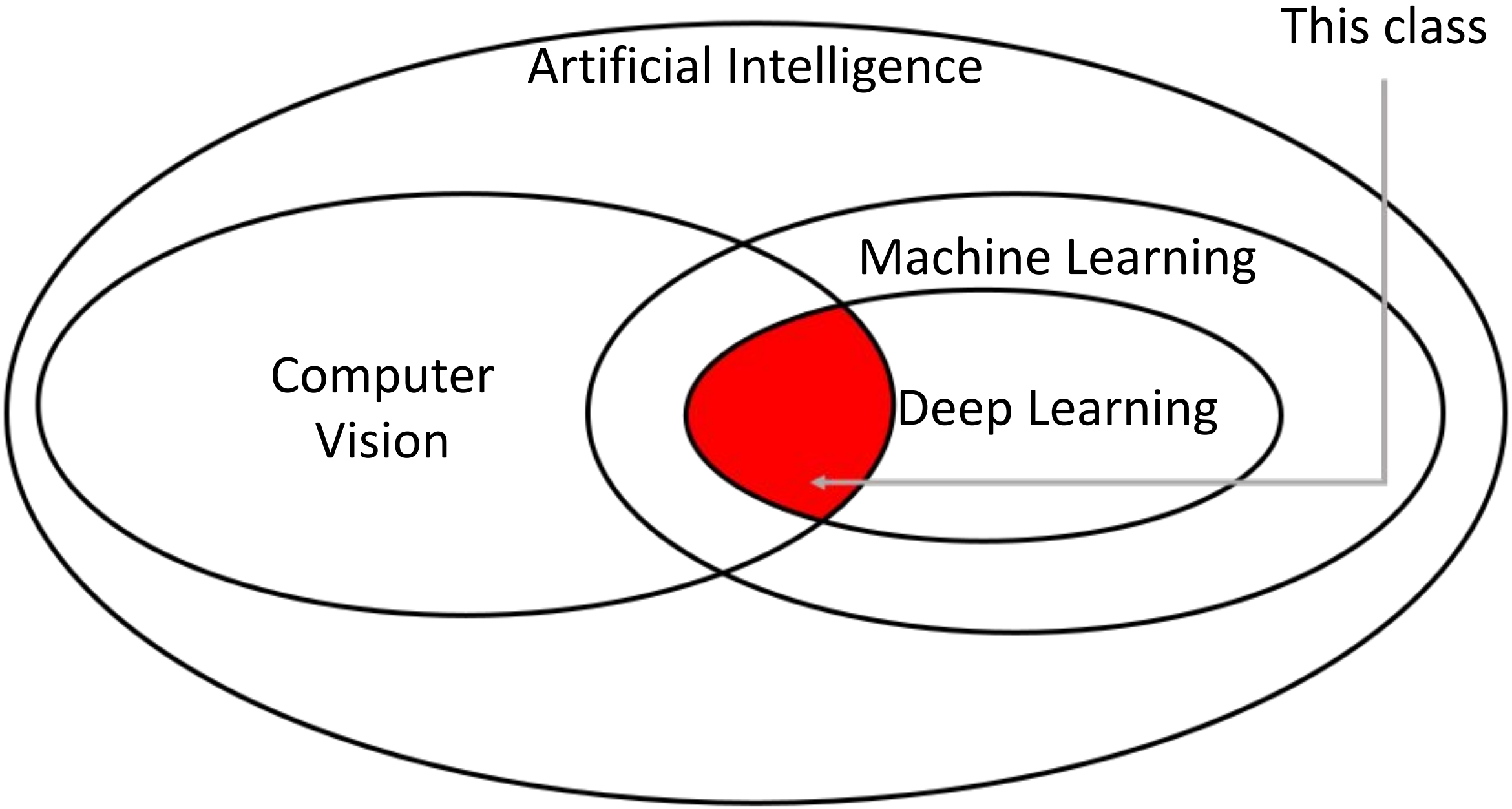
Deep Learning For Computer Vision

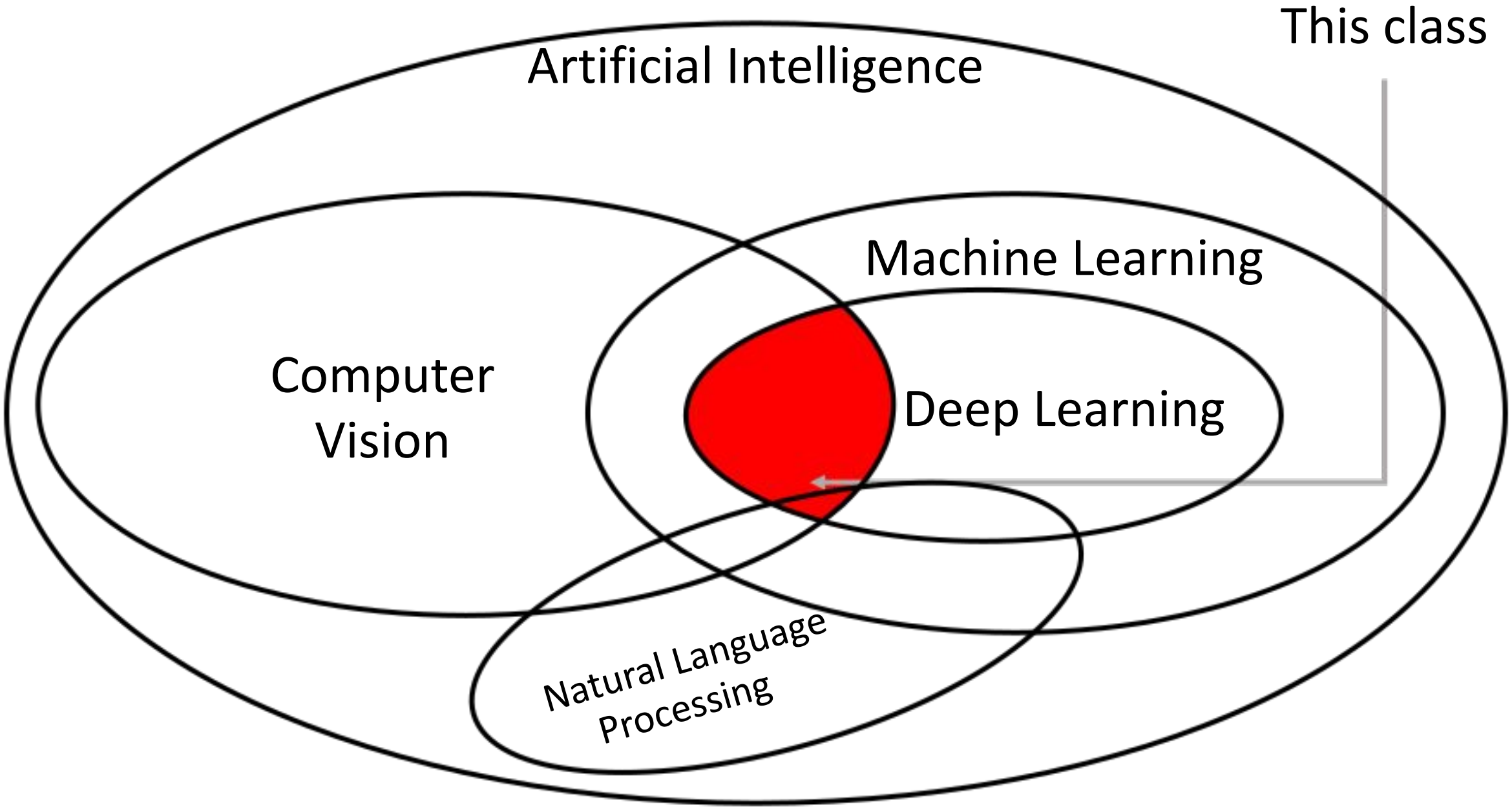
Hierarchical learning algorithms with many “layers”, (very) loosely inspired by the brain

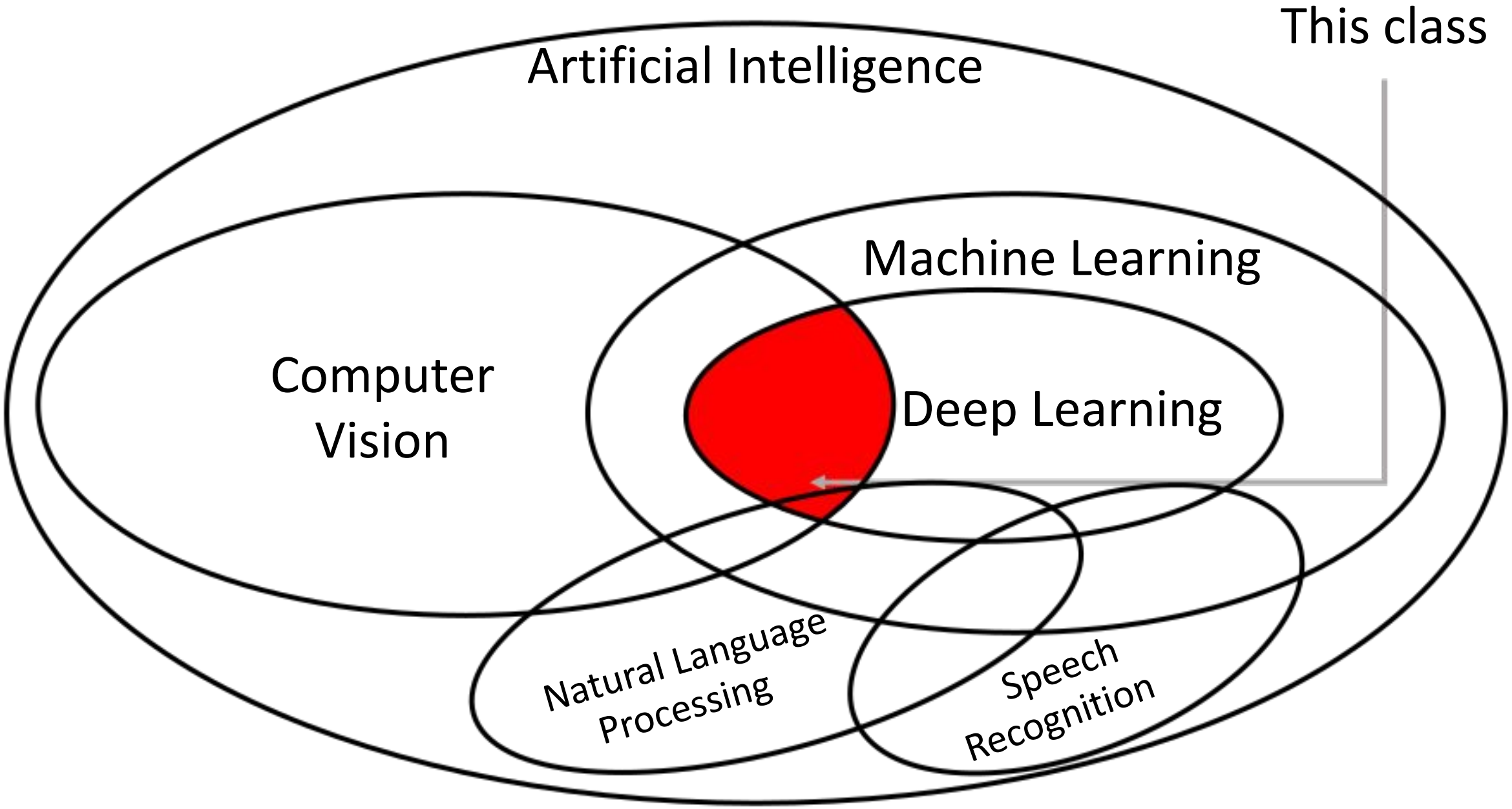

Artificial Intelligence

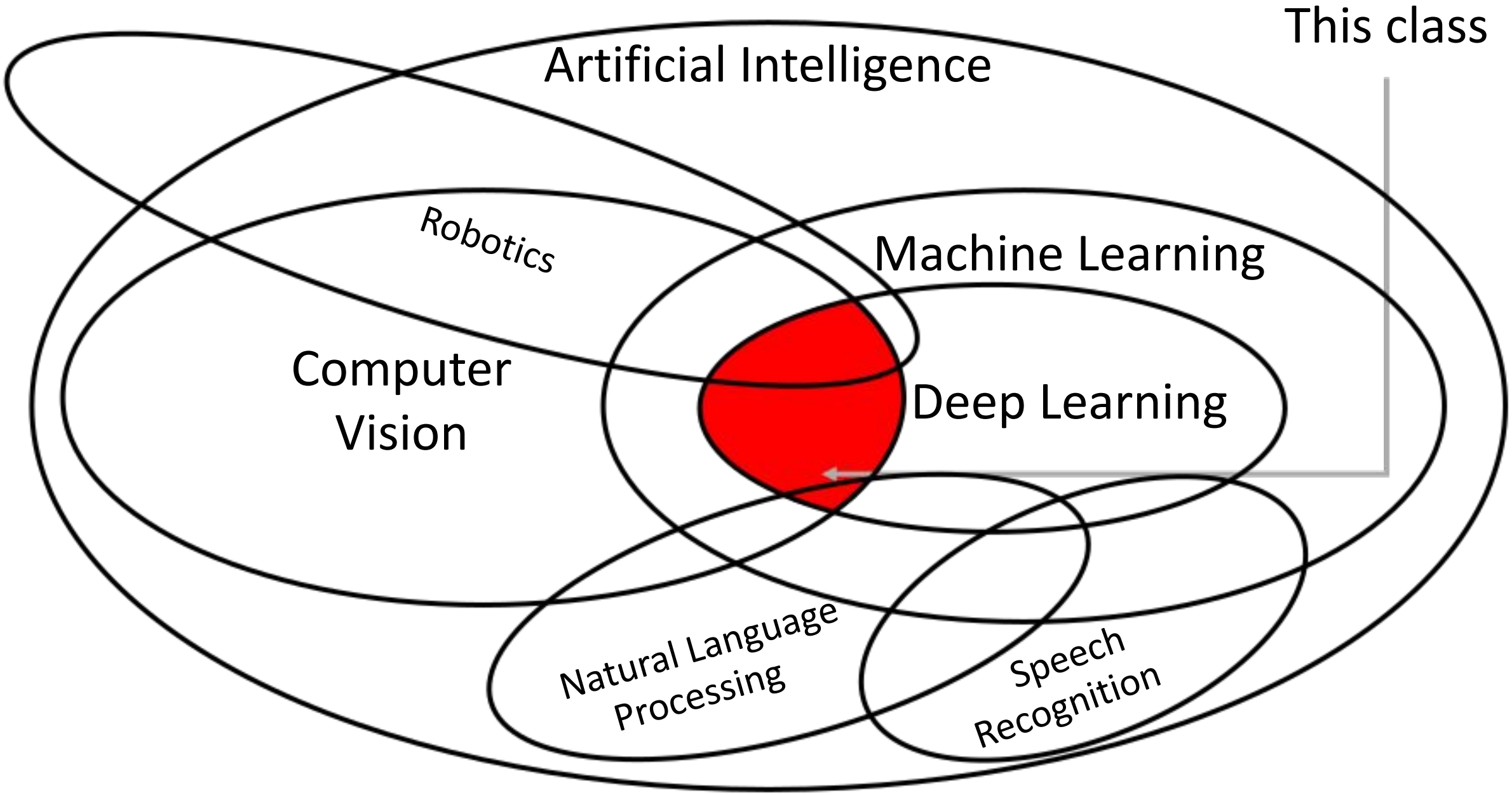

Artificial Intelligence
Machine Learning
Computer
Vision


Artificial Intelligence
Machine Learning
Computer
Vision
Deep Learning


This class
Artificial Intelligence
Machine Learning
Computer
Vision
Deep Learning


This class
Artificial Intelligence
Machine Learning
Computer Vision
Deep Learning
Natural Language Processing


This class
Artificial Intelligence
Machine Learning
Computer Vision
Deep Learning
Natural Language Processing
Speech Recognition


This class
Artificial Intelligence
Robotics
Machine Learning
Computer Vision
Deep Learning
Natural Language Processing
Speech Recognition

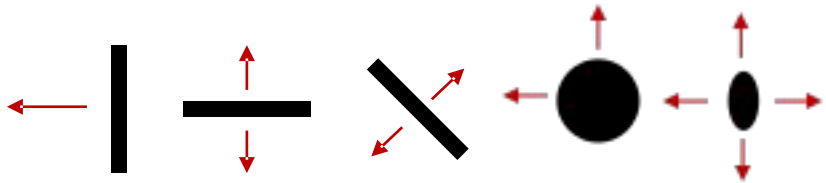
Today’s Agenda

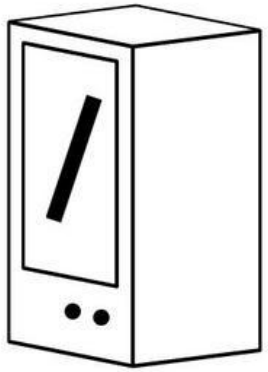
- A brief history of computer vision and deep learning
- Course overview and logistics

Hubel and Wiesel, 1959

Measure brain activity

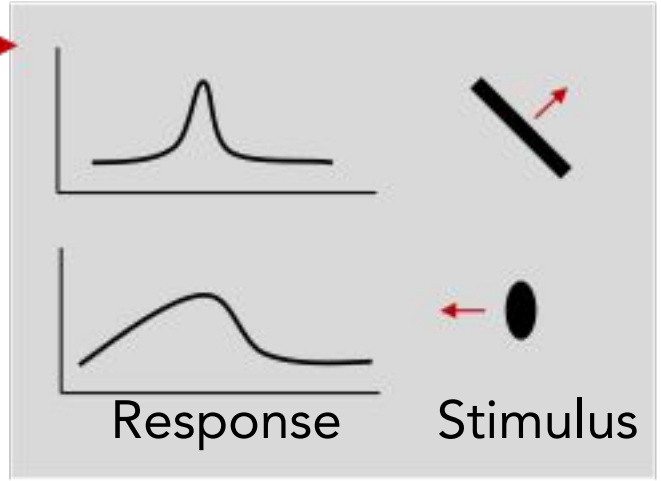






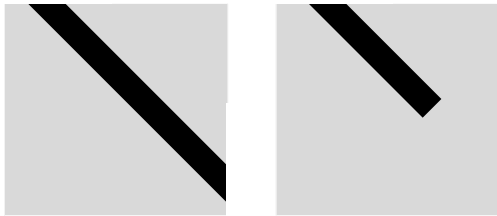
Simple cells:

Response to light orientation



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1959 Hubel & Wiesel

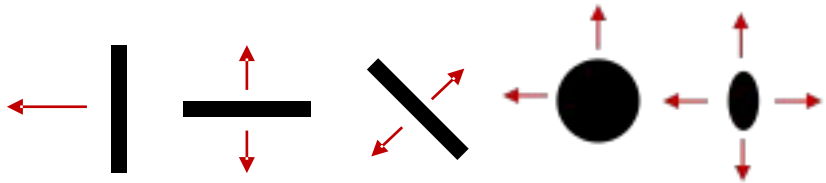


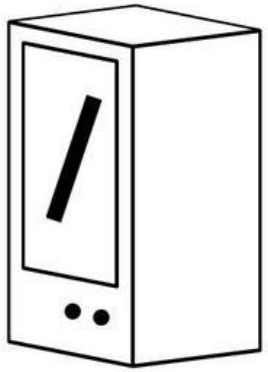
No response

Hubel and Wiesel, 1959

Measure brain activity





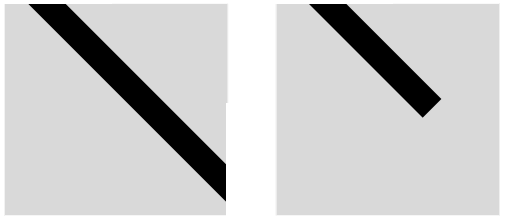


Simple cells:

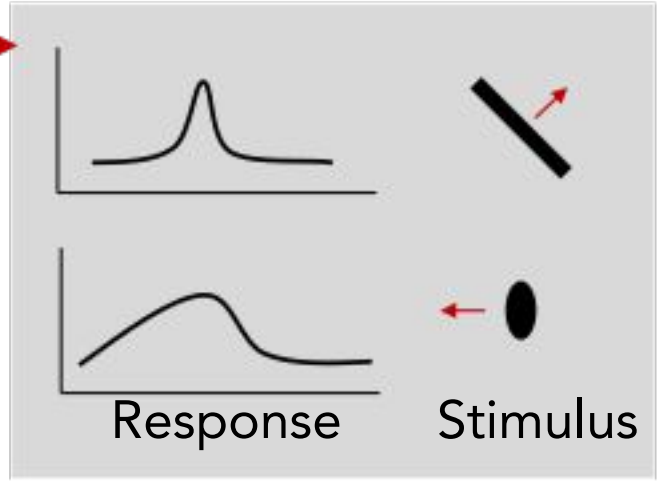
Response to light orientation

Complex cells:

Response to light orientation and movement



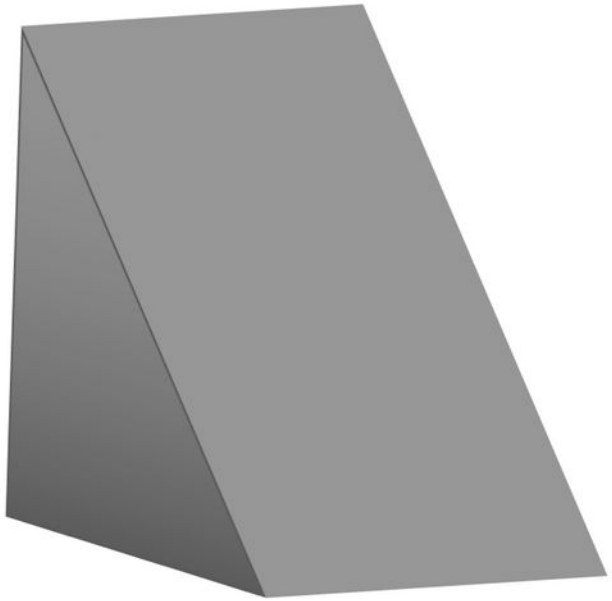
No response



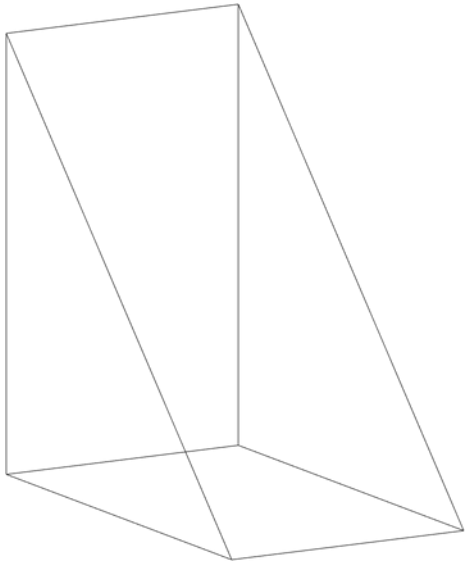
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1959 Hubel & Wiesel

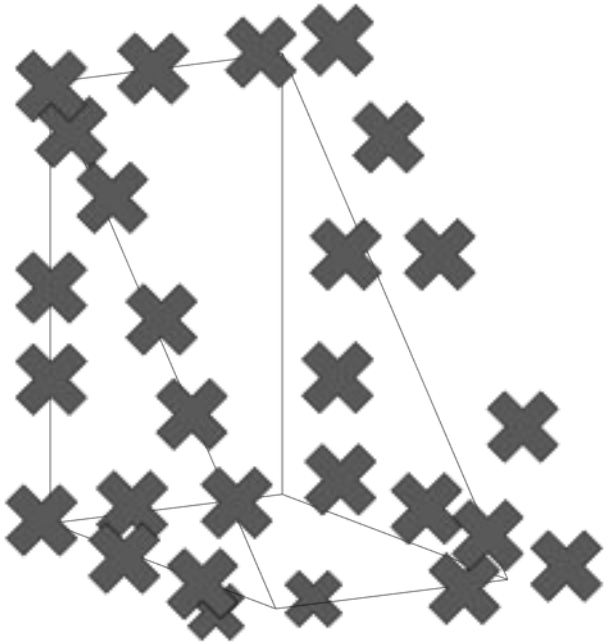
Larry Roberts, 1963



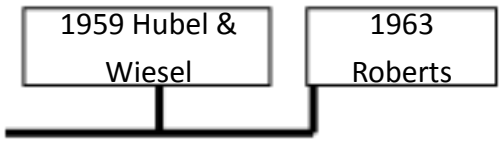
(a) Original picture



(b) Differentiated picture



(c) Feature points selected



Lawrence Gilman Roberts, “Machine Perception of Three-Dimensional Solids”, 1963

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

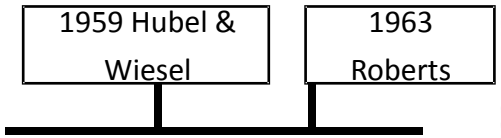
PROJECT MAC

Artificial Intelligence Group — July 7, 1966
Vision Memo. No. 100.

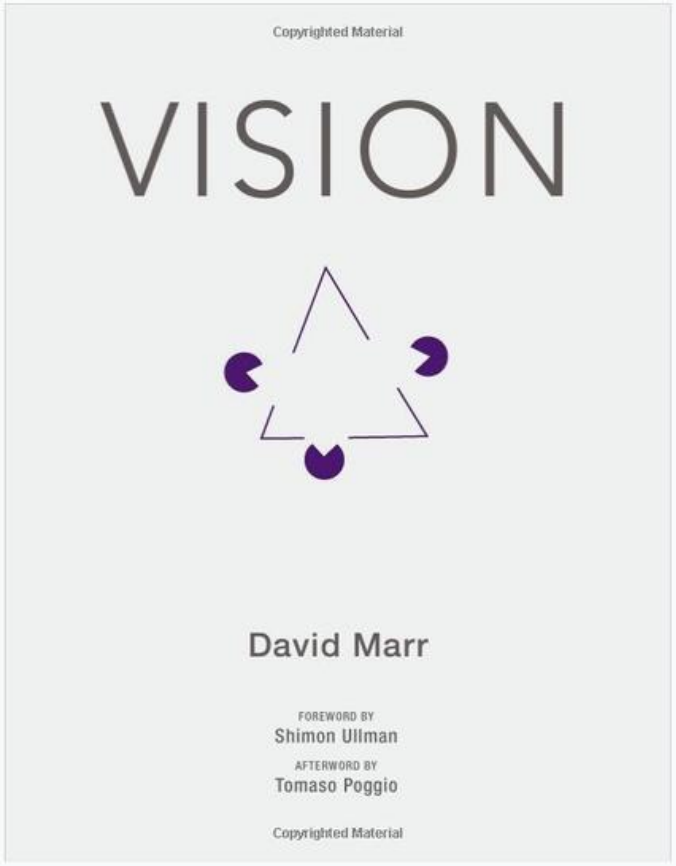
THE SUMMER VISION PROJECT

Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".



https://dspace.mit.edu/handle/1721.1/6125



1959 Hubel & Wiesel

1963 Roberts

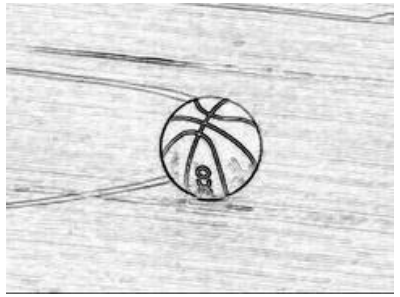
1970s David Marr

Input image

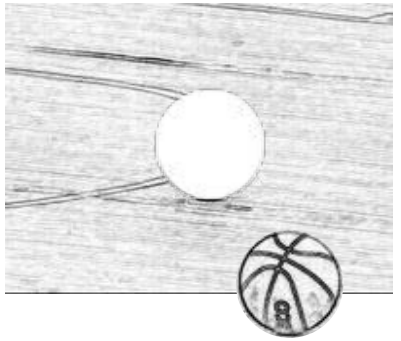


This image is CC0 1.0 public domain

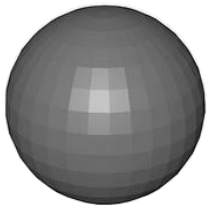
Edge image



2 ½-D sketch



3-D model

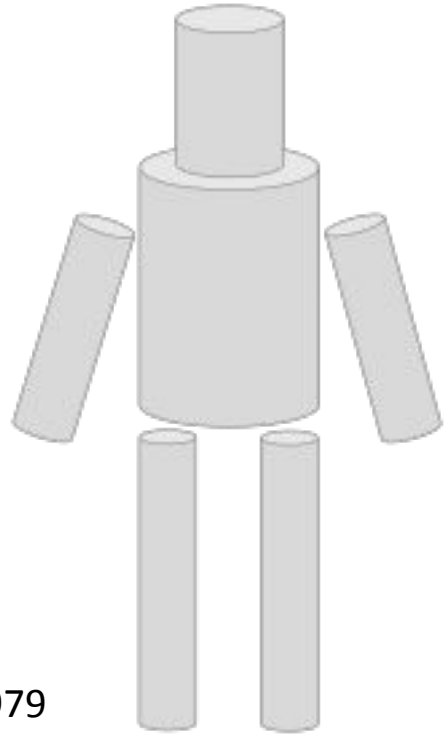


This image is CC0 1.0 public domain

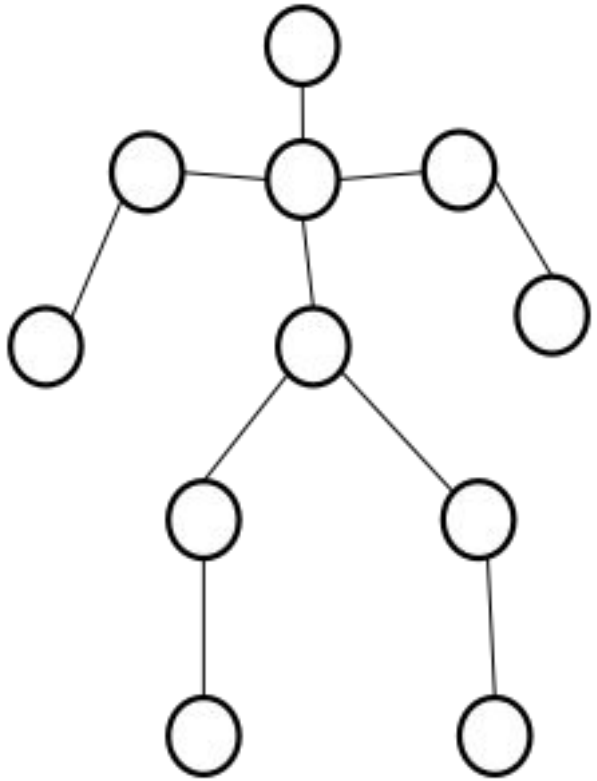
Input Image	Primal Sketch	2 ½-D Sketch	3-D Model Representation
Perceived intensities	Zero crossings, blobs, edges, bars, ends, virtual lines, groups, curves boundaries	Local surface orientation and discontinuities in depth and in surface orientation	3-D models hierarchically organized in terms of surface and volumetric primitives

Stages of Visual Representation, David Marr, 1970s

Recognition via Parts (1970s)



Generalized Cylinders,
Brooks and Binford, 1979



Pictorial Structures,
Fischlerand Elshlager, 1973

1959 Hubel & Wiesel | 1963 Roberts | 1970s David Marr | 1979 Gen. Cylinders

Recognition via Edge Detection (1980s)





1959 Hubel & Wiesel

1963 Roberts

1970s David Marr

1979 Gen. Cylinders

1986 Canny

John Canny, 1986

David Lowe, 1987

Image is CC0 1.0 public domain

Recognition via Grouping (1990s)



1959 Hubel & Wiesel

1963 Roberts

1970s David Marr

1979 Gen. Cylinders

1986 Canny

AI Winter

1997 Norm. Cuts

Normalized Cuts, Shi and Malik, 1997

Left Image is CC BY 3.0 Middl Image is public domain Right Image is CC-BY 2.0; changes made

Recognition via Matching (2000s)



Imageis public domain



Imageis public domain

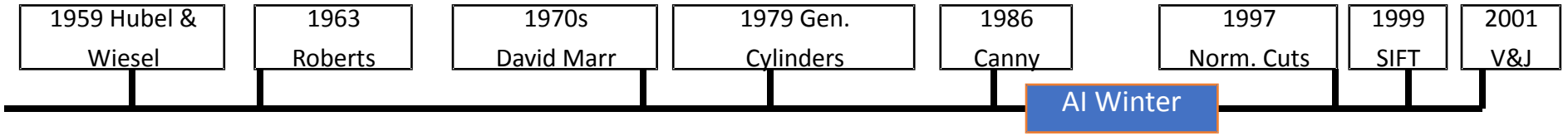
1959 Hubel & Wiesel | 1963 Roberts | 1970s David Marr | 1979 Gen. Cylinders | 1986 Canny | AI Winter | 1997 Norm. Cuts | 1999 SIFT

SIFT, David Lowe, 1999

Face Detection

Viola and Jones, 2001 One of the first successful applications of machine learning to vision





PASCAL Visual Object Challenge

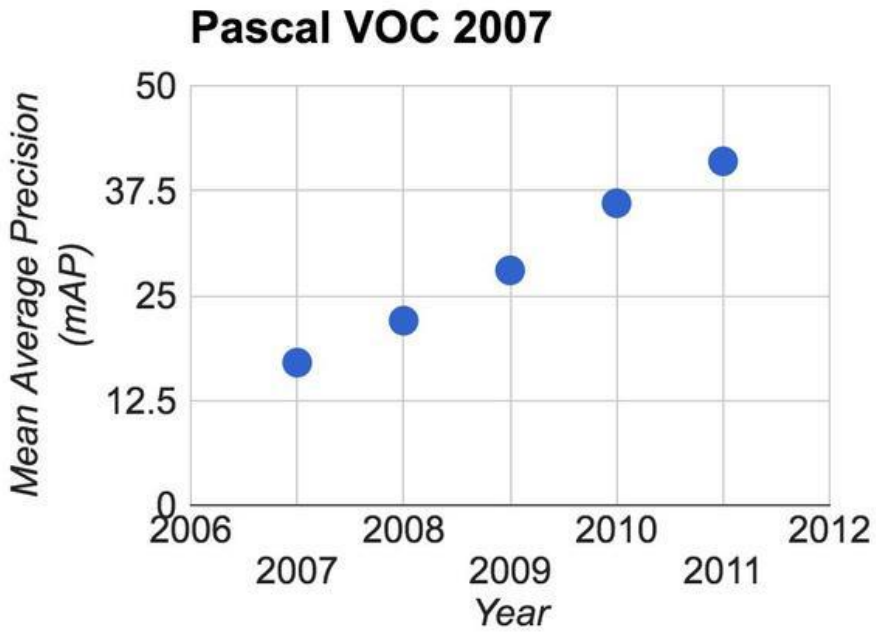
Imageis CC0 1.0public domain







Imageis CC0 1.0public domain



1959 Hubel & Wiesel
1963 Roberts
1970s David Marr
1979 Gen. Cylinders
1986 Canny
AI Winter
1997 Norm. Cuts
1999 SIFT
2001 V&J
2007 PASCAL

IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:

1,000 object classes 1,431,167 images



Output:

Scale

T-shirt

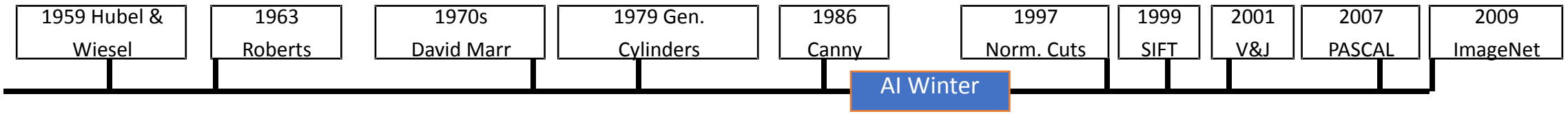
Steel drum

Drumstick

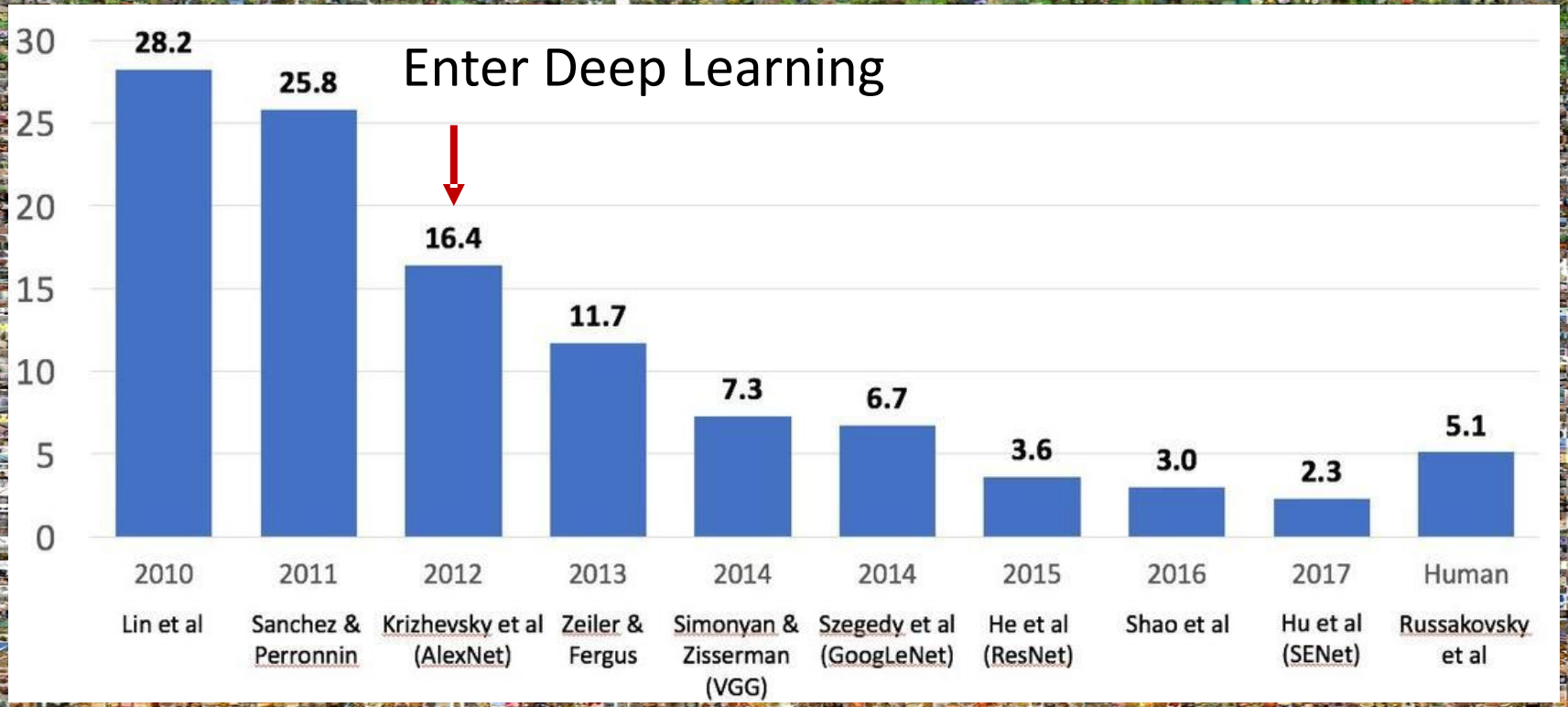
Mud turtle

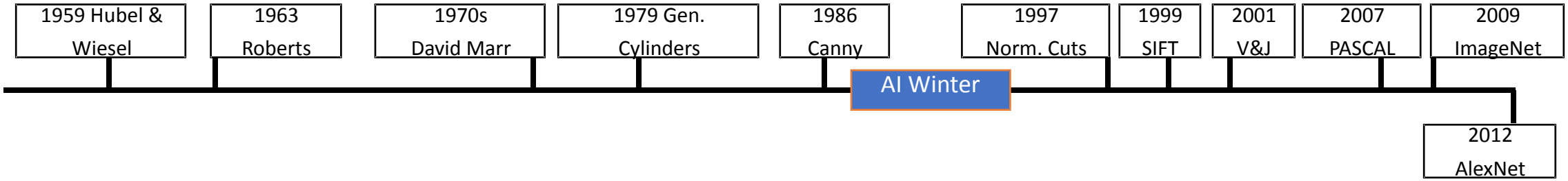
Deng et al, 2009

Russakovskyet al. IJCV 2015

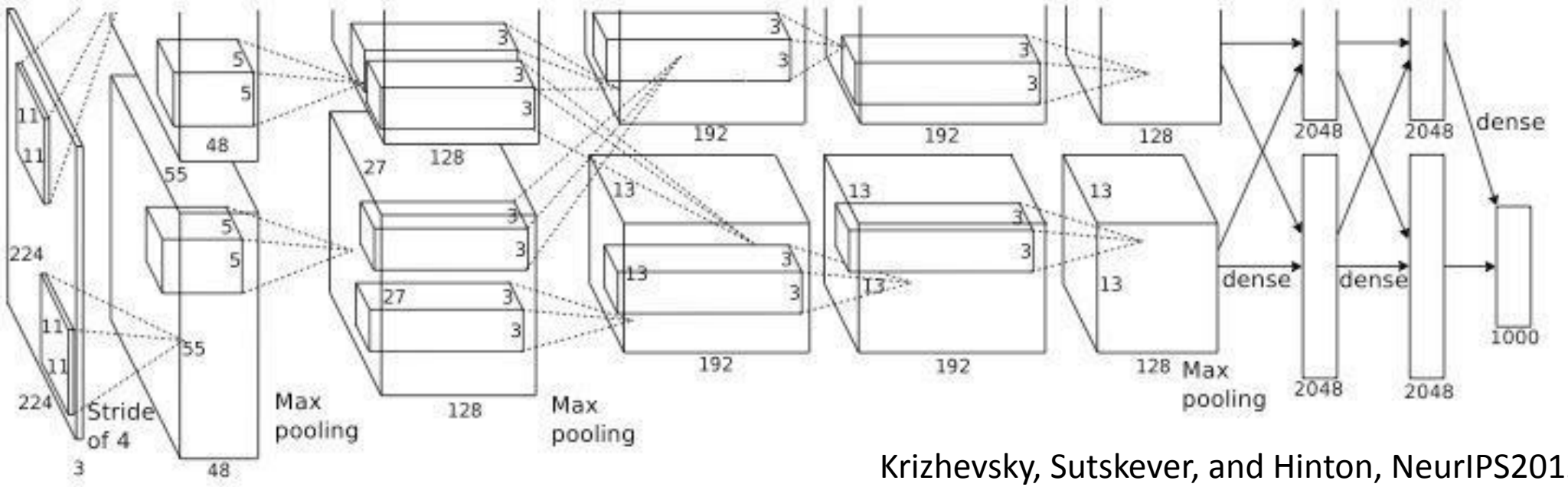


IMAGENET Large Scale Visual Recognition Challenge

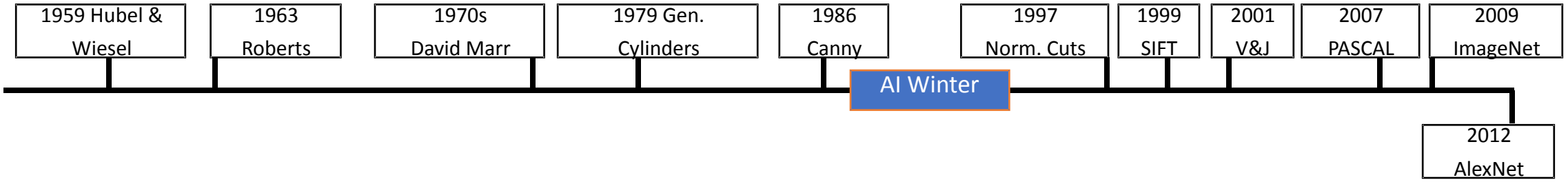




AlexNet: Deep Learning Goes Mainstream



Krizhevsky, Sutskever, and Hinton, NeurIPS2012



Perceptron

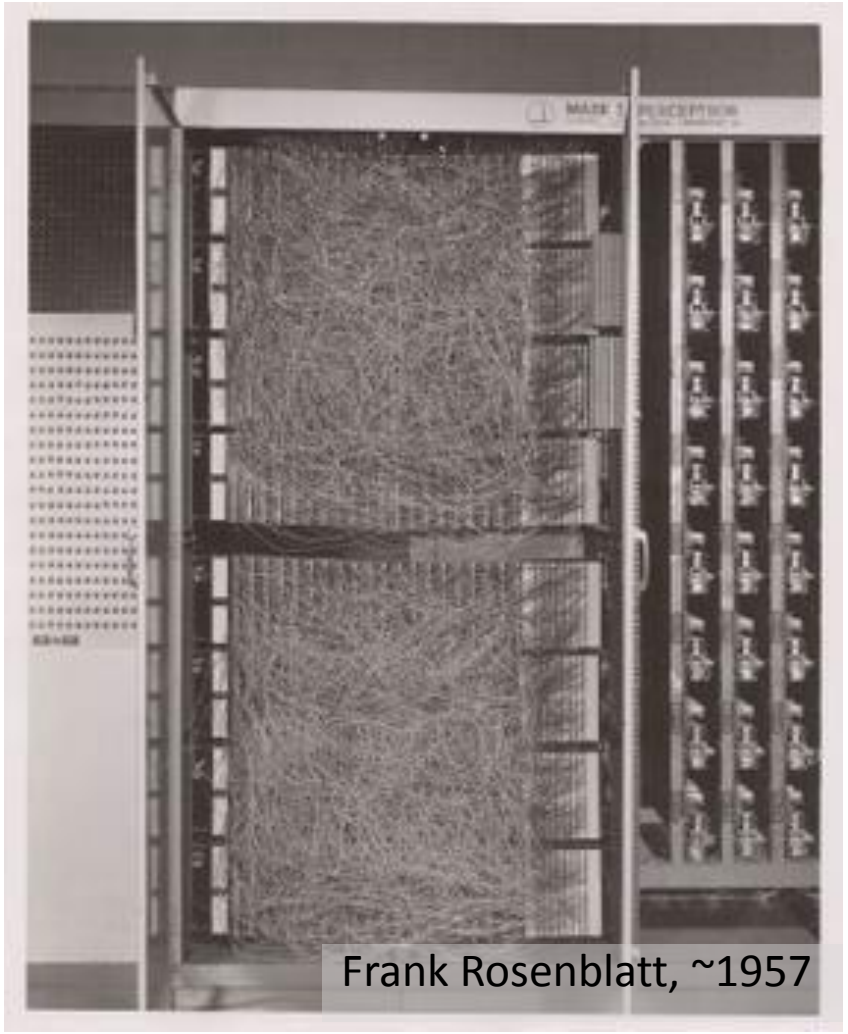
One of the earliest algorithms that could learn from data

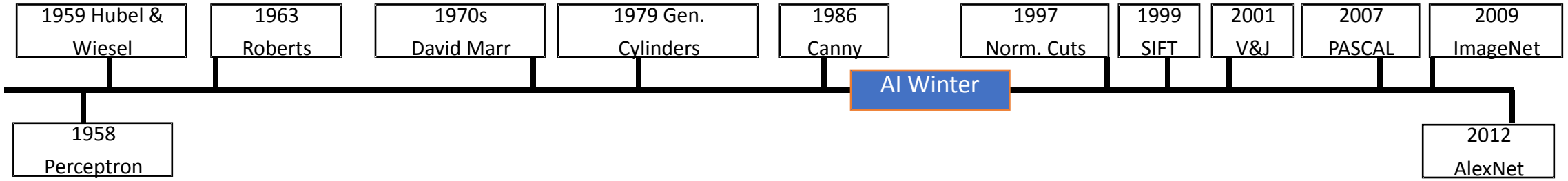
Implemented in hardware! Weights stored in potentiometers, updated with electric motors during learning

Connected to a camera that used 20x20 cadmium sulfide photocells to make a 400-pixel image

Could learn to recognize letters of the alphabet

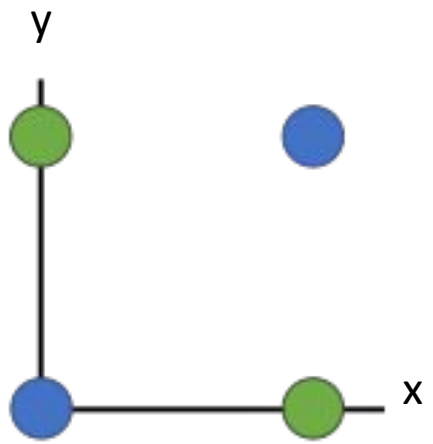
Today we would recognize it as a **linear classifier**

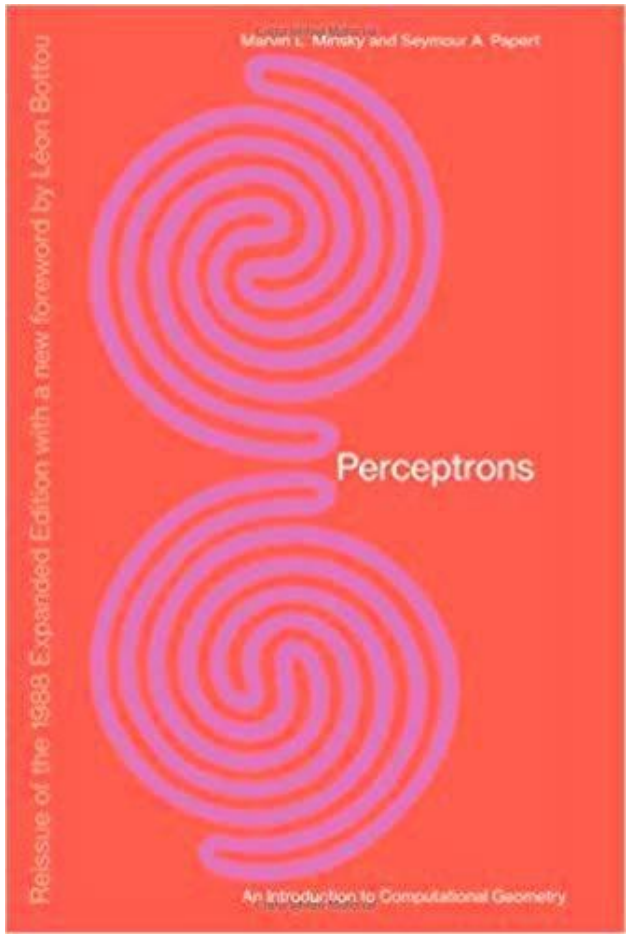

Frank Rosenblatt, ~1957



Minsky and Papert, 1969

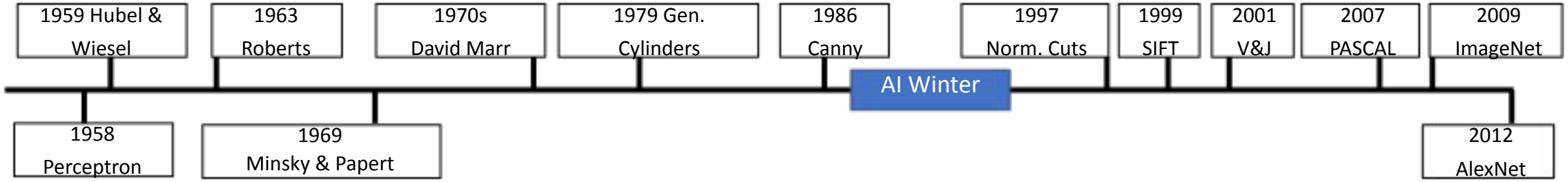
X	Y	F(x,y)
0	0	0
0	1	1
1	0	1
1	1	0





Showed that Perceptrons could not learn the XOR function

Caused a lot of disillusionment in the field

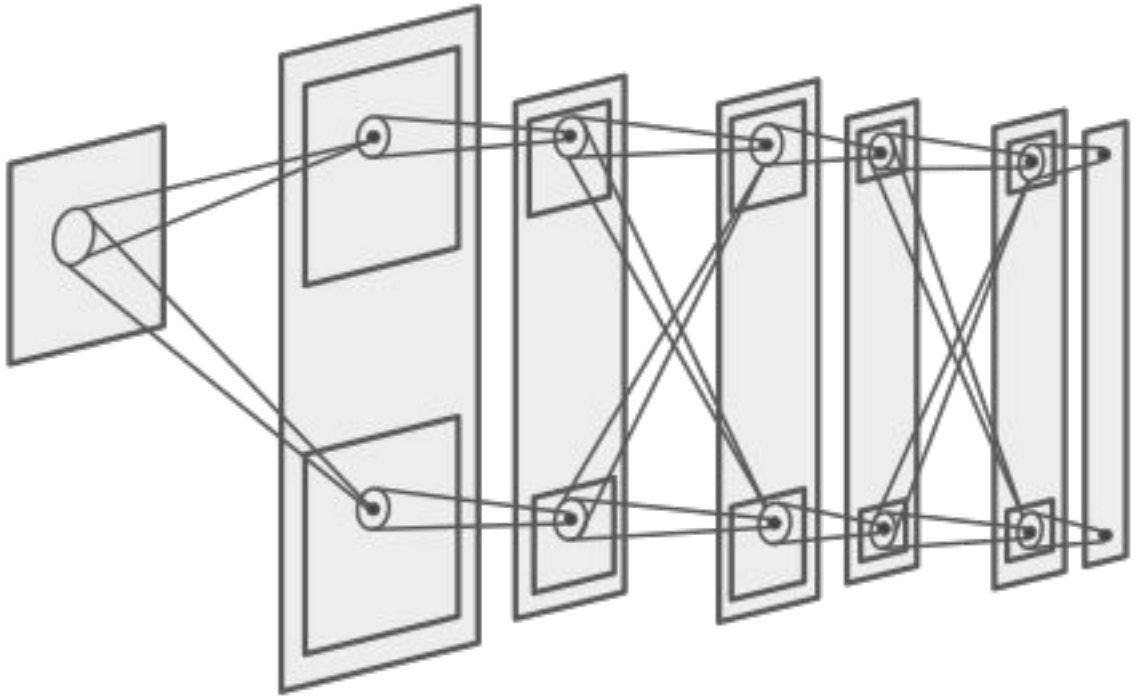


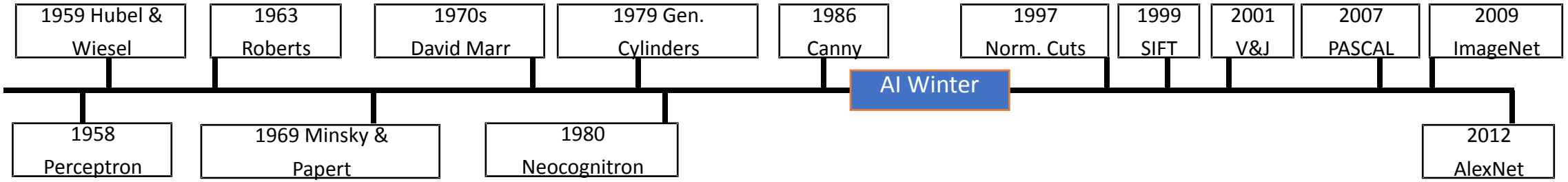
Neocognitron: Fukushima, 1980

Computational model the visual system, directly inspired by Hubel and Wiesel's hierarchy of complex and simple cells

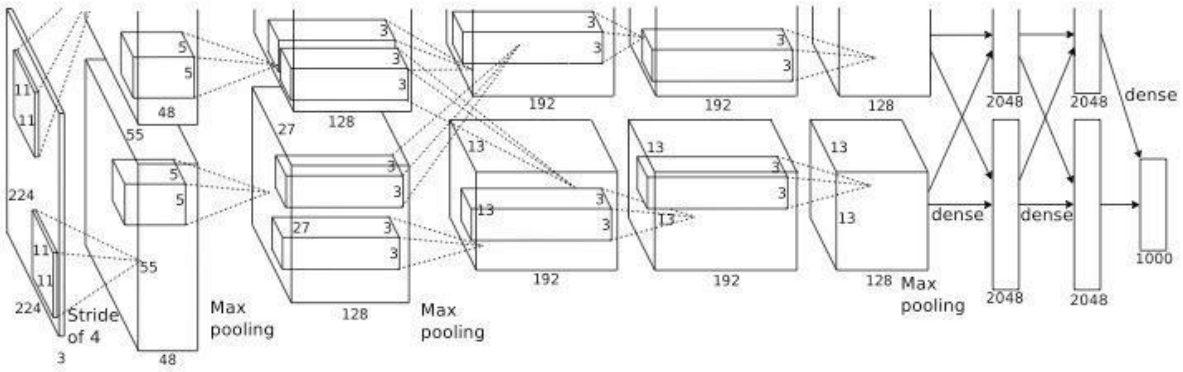
Interleaved simple cells (convolution) and complex cells (pooling)

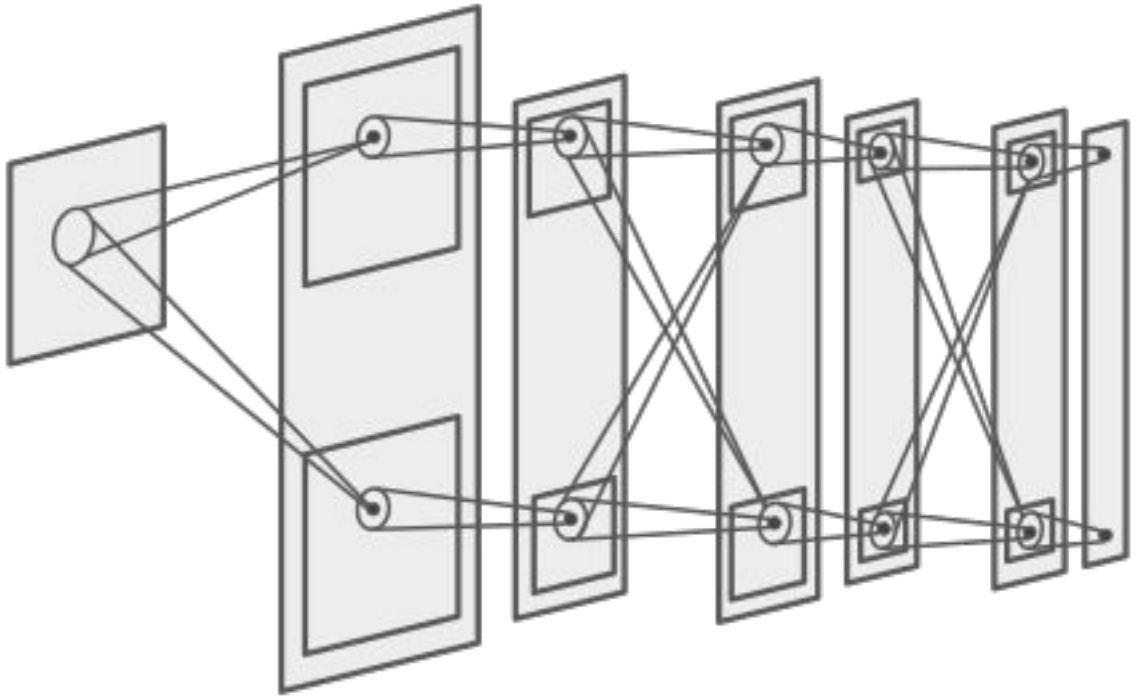
No practical training algorithm





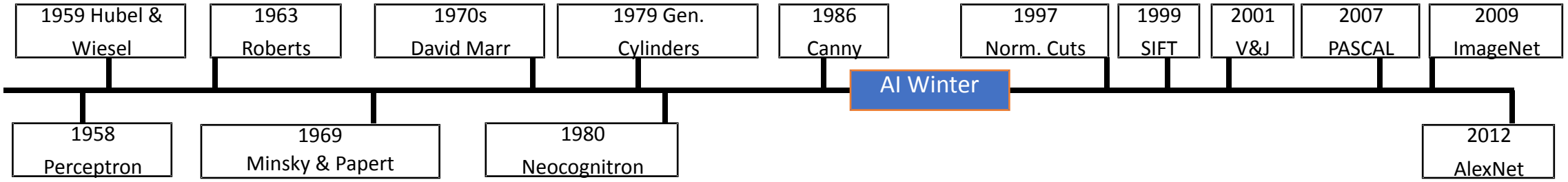
Neocognitron: Fukushima, 1980





Looks a lot like AlexNet

more than 32 years later!



Backprop: Rumelhart, Hinton, and Williams, 1986

Introduced backpropagation for computing gradients in neural networks

Successfully trained perceptrons with multiple layers

$$\frac{\partial E_p}{\partial w_{ji}} = \frac{\partial E_p}{\partial o_{pj}} \frac{\partial o_{pj}}{\partial w_{ji}}$$

recognizable math

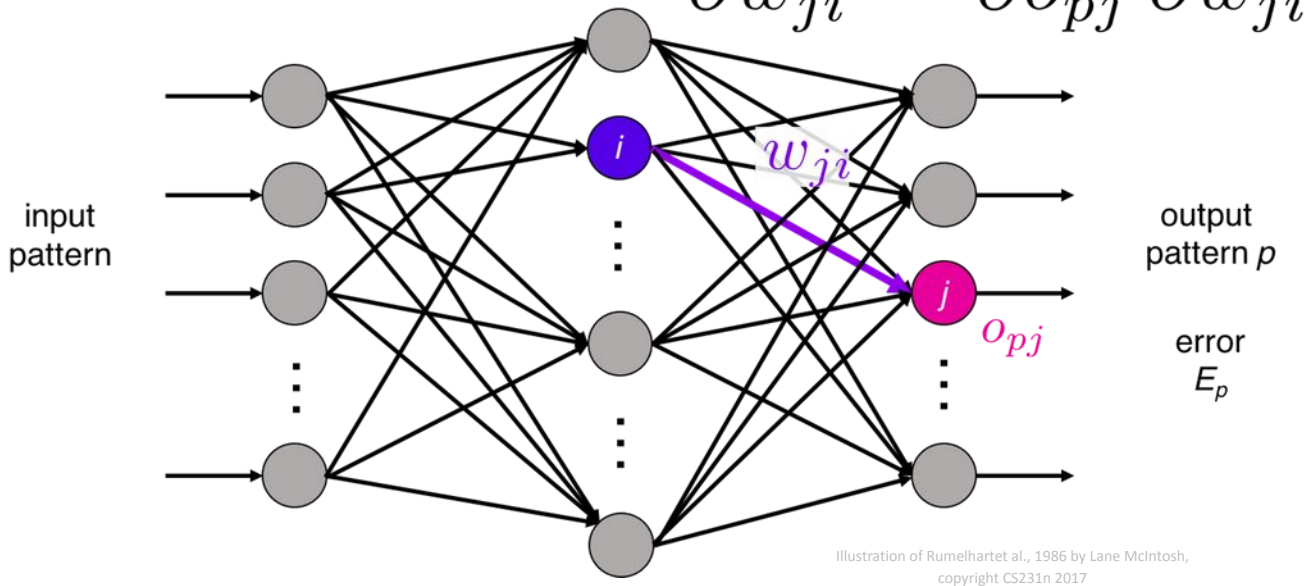
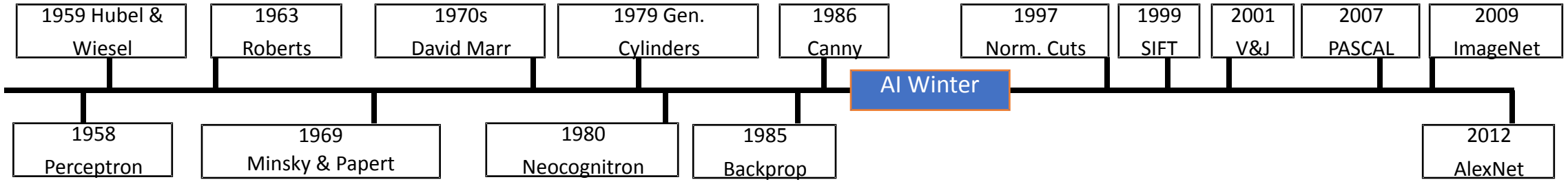
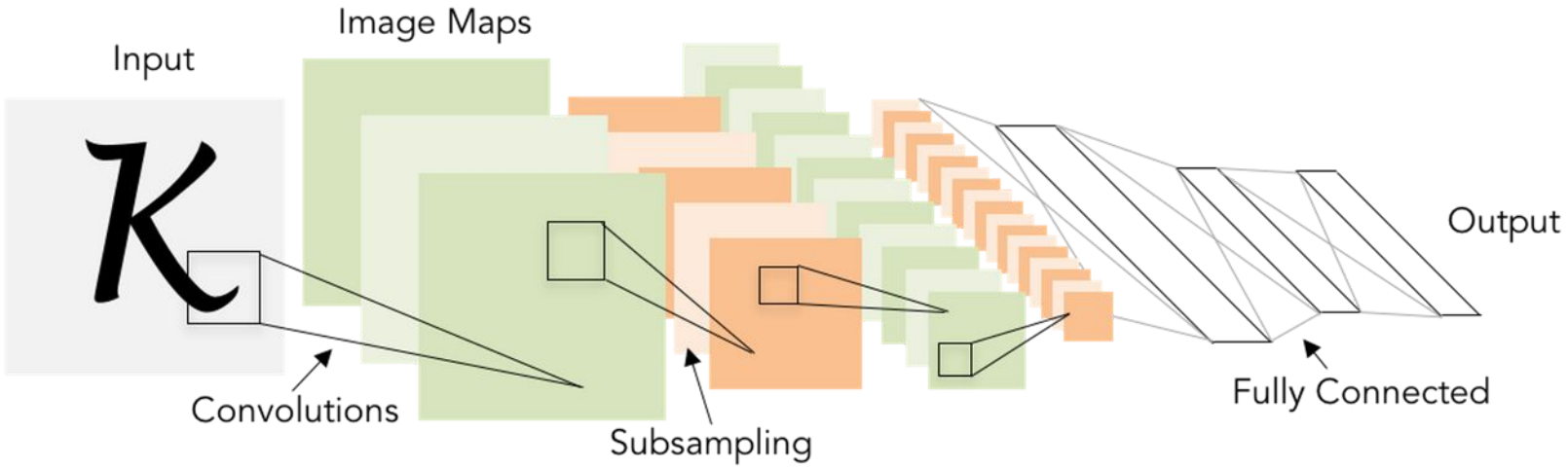


Illustration of Rumelhartet al., 1986 by Lane McIntosh, copyright CS231n 2017



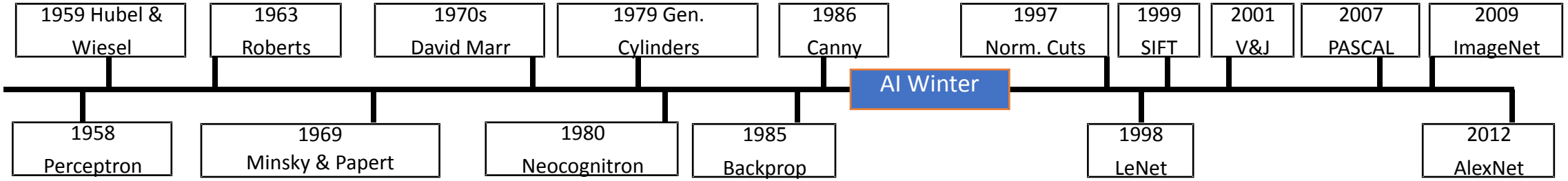
Convolutional Networks: LeCun et al, 1998



Applied backprop algorithm to a Neocognitron-like architecture Learned to recognize handwritten digits

Was deployed in a commercial system by NEC, processed handwritten checks

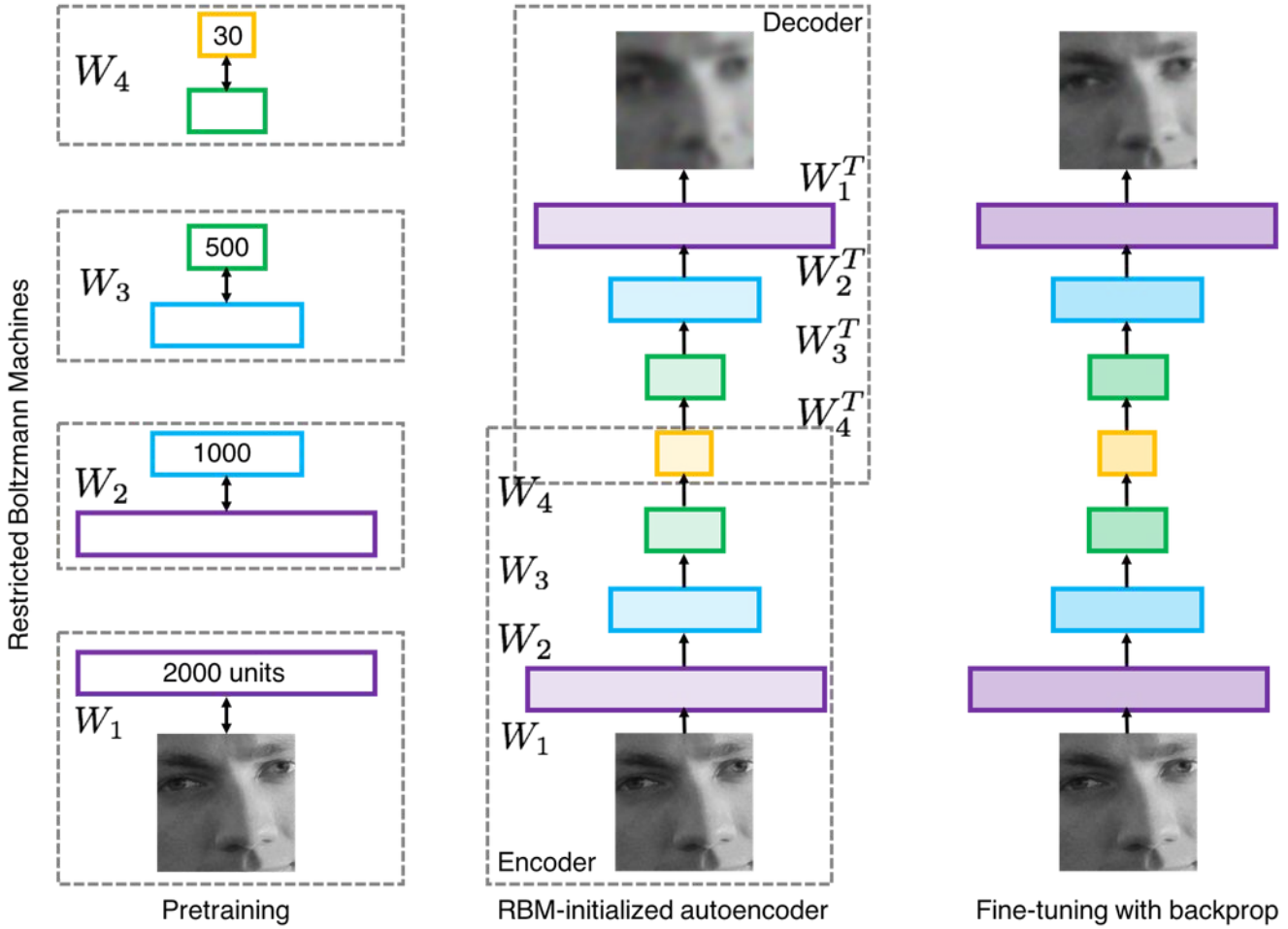
Very similar to our modern convolutional networks!



2000s: “Deep Learning”

People tried to train neural networks that were deeper and deeper

Not a mainstream research topic at this time

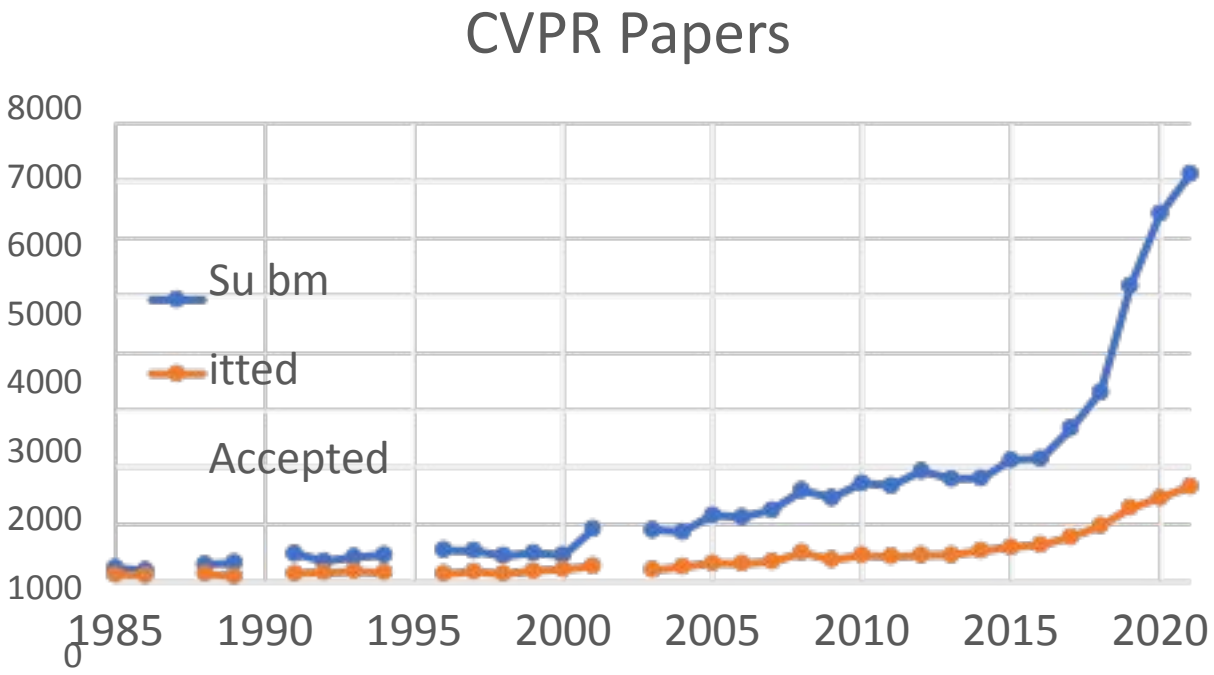


1959 Hubel & Wiesel
1963 Roberts
1970s David Marr
1979 Gen. Cylinders
1986 Canny
1997 Norm. Cuts
1999 SIFT
2001 V&J
2007 PASCAL
2009 ImageNet

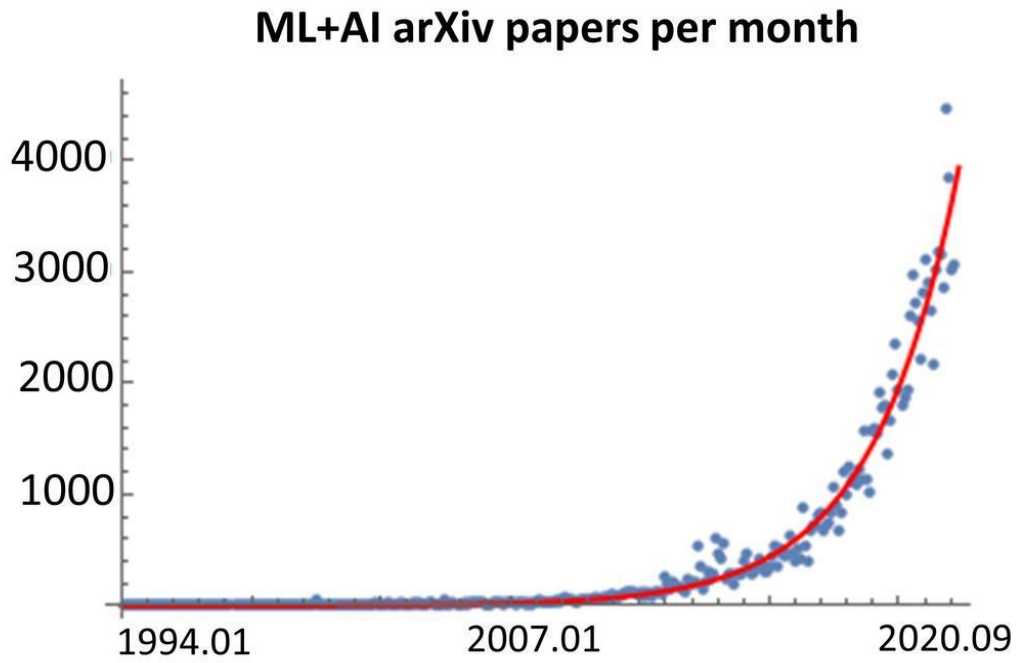
AI Winter

1958 Perceptron
1969 Minsky & Papert
1980 Neocognitron
1985 Backprop
1998 LeNet
2006 Deep Learning
2012 AlexNet

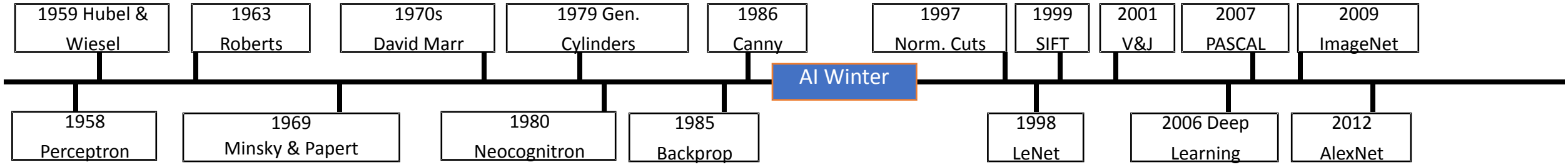
2012 to Present: Deep Learning Explosion



Publications at top Computer Vision conference



arXiv papers per month (source)



2012 to Present: Deep Learning is Everywhere

Image Classification

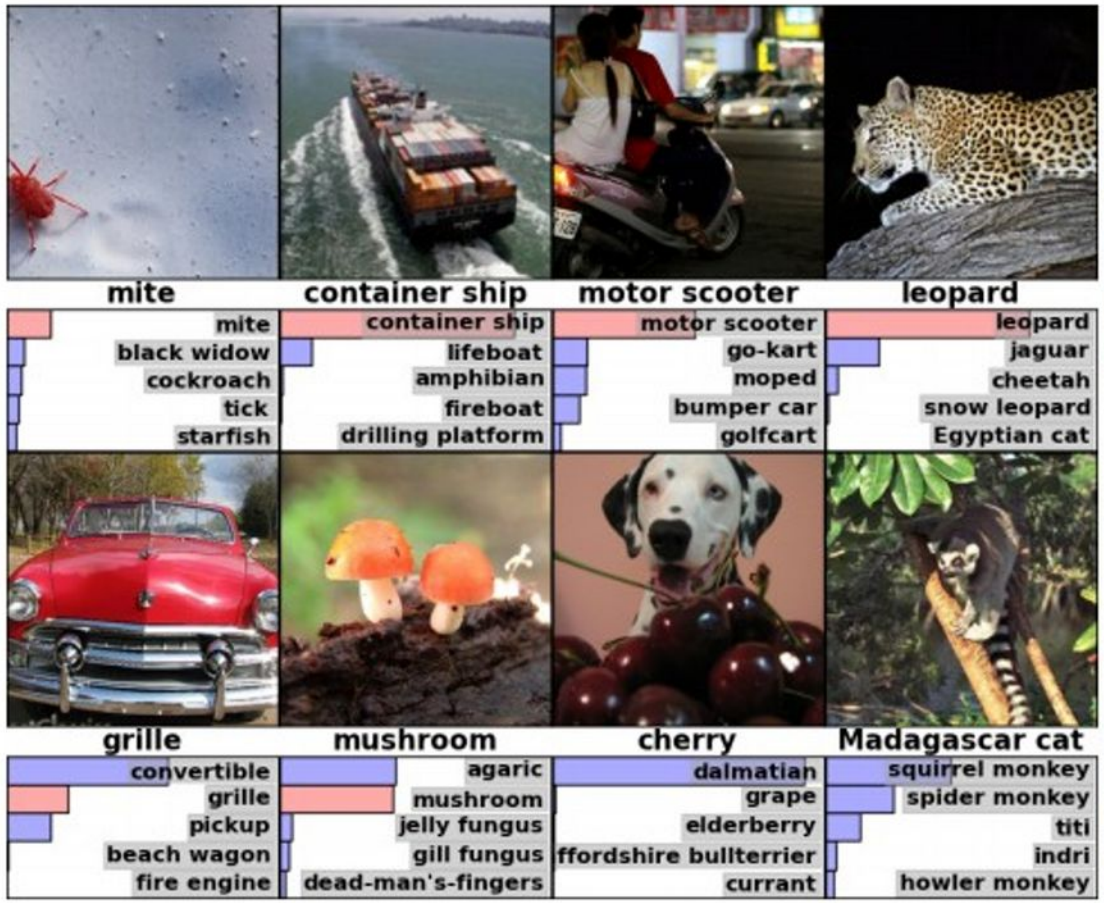


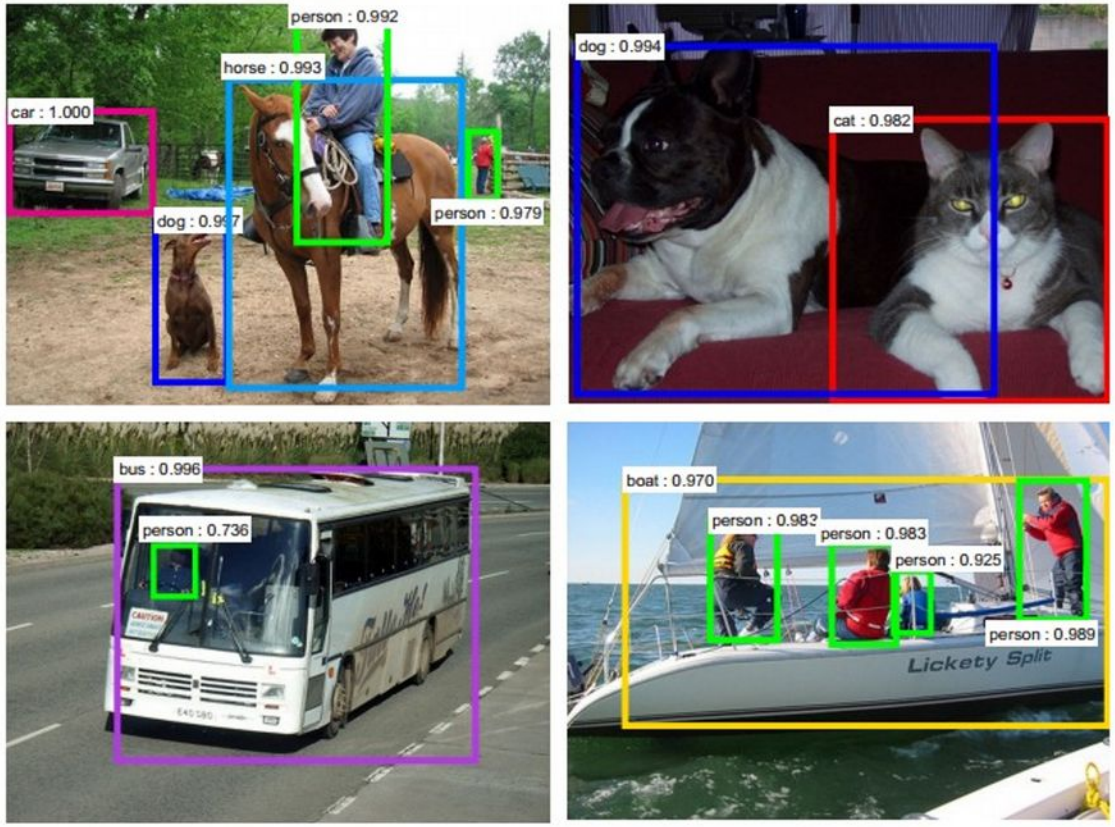
Image Retrieval



Figures copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

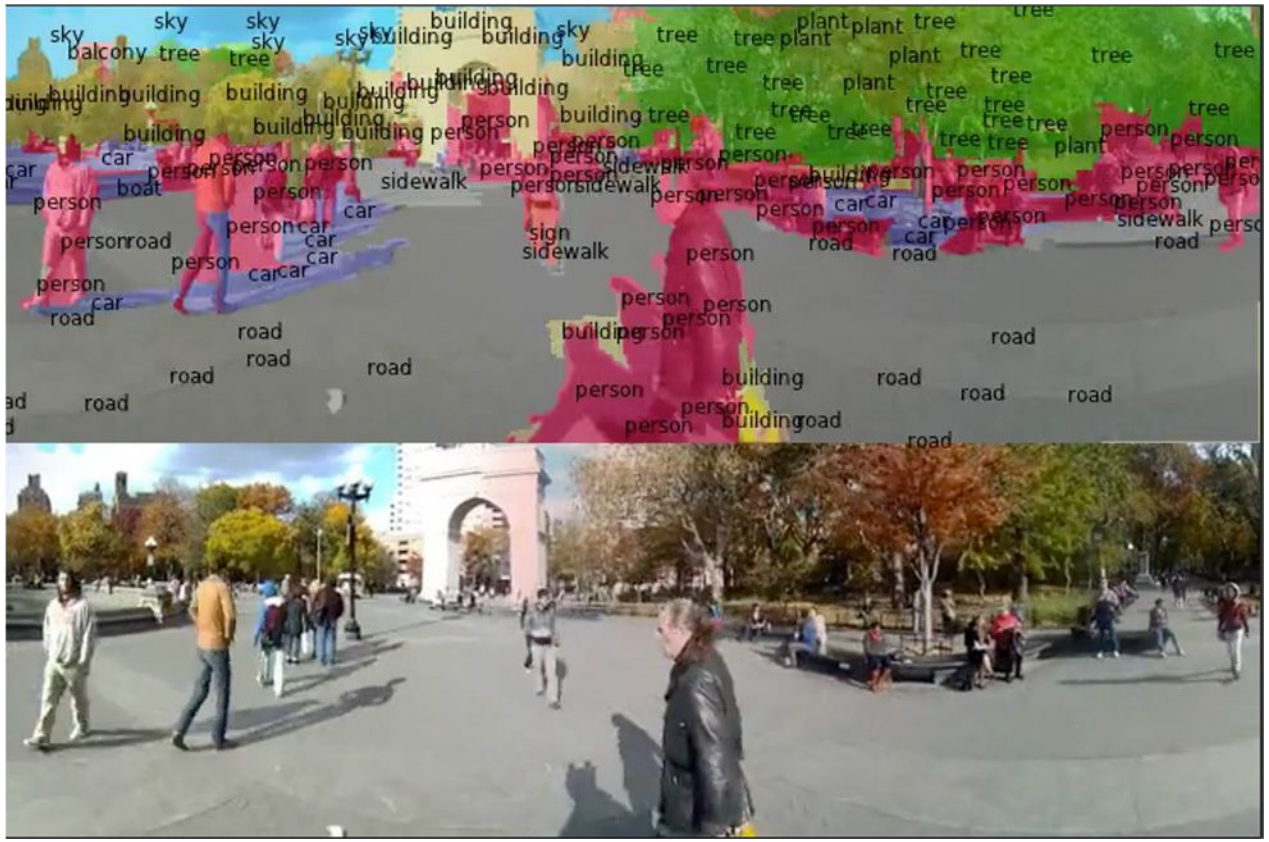
2012 to Present: Deep Learning is Everywhere

Object Detection



Ren, He, Girshick, and Sun, 2015

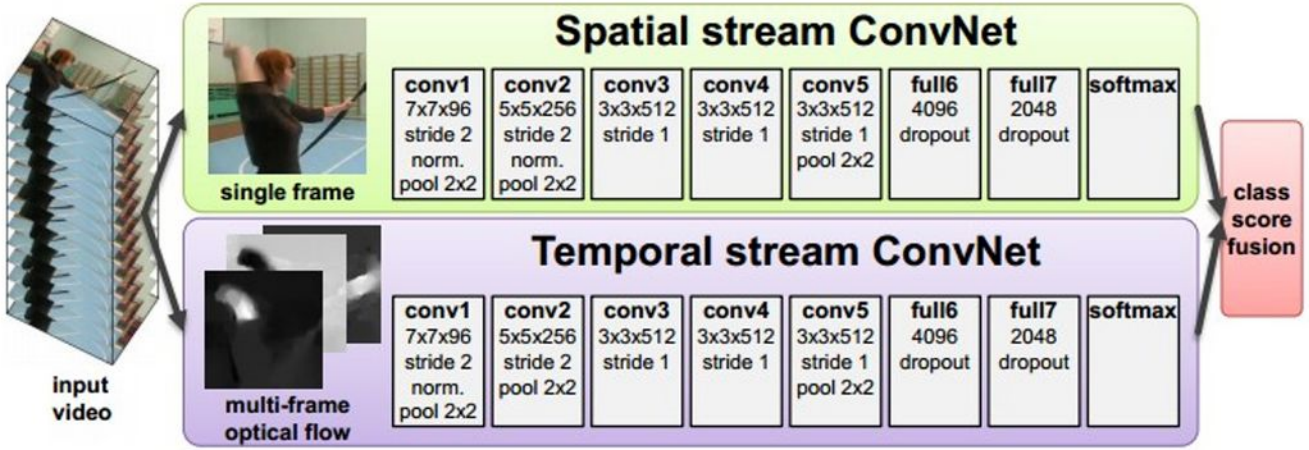
Image Segmentation



Fabaretet al, 2012

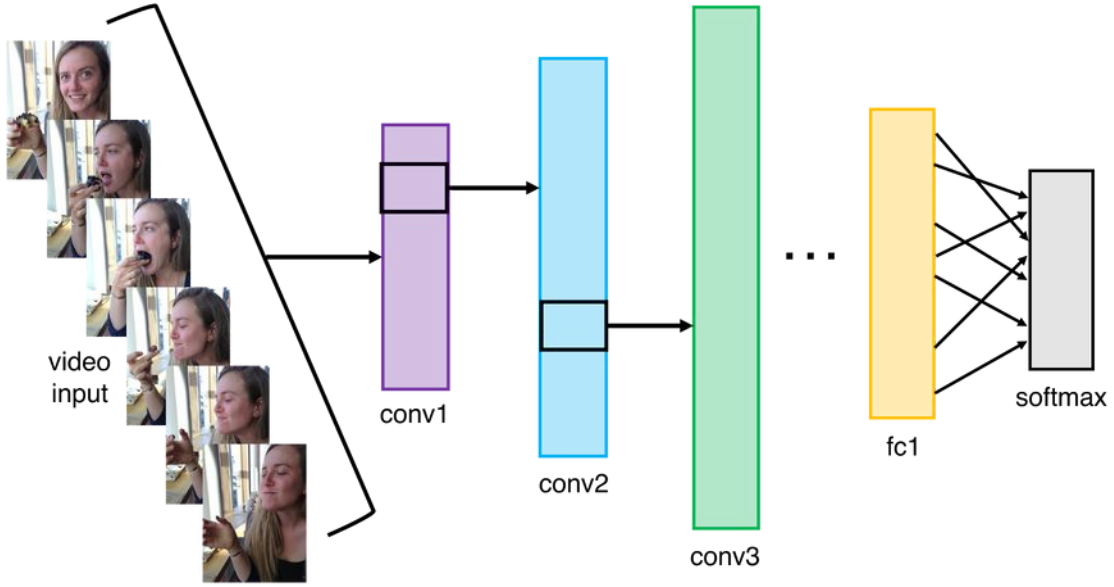
2012 to Present: Deep Learning is Everywhere

Video Classification



Simonyanet al, 2014

Activity Recognition

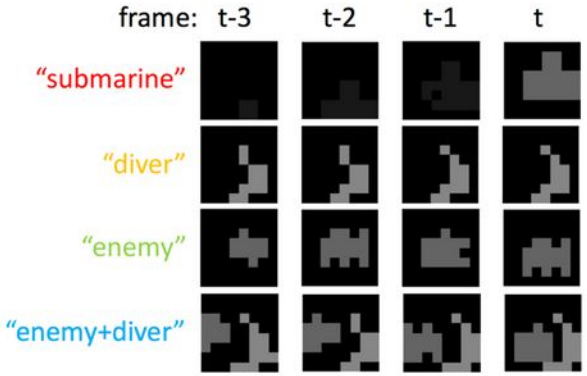


2012 to Present: Deep Learning is Everywhere

Pose Recognition (Toshevand Szegedy, 2014)

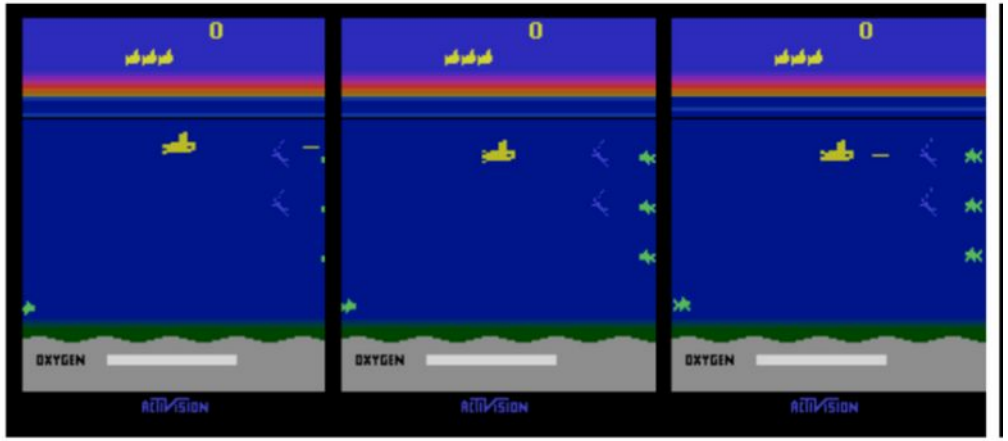


Playing Atari games (Guo et al, 2014)



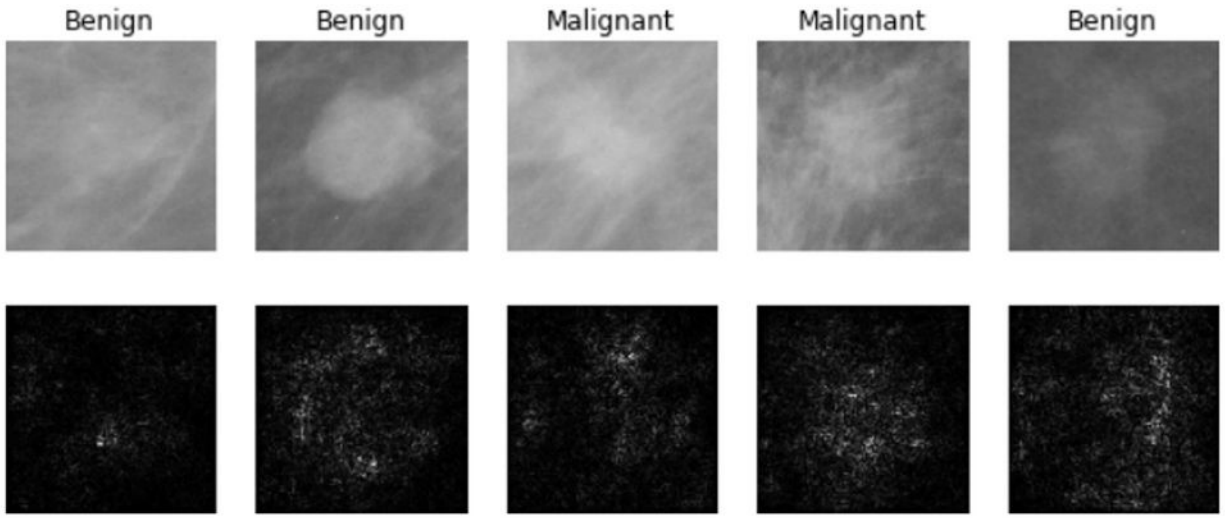






2012 to Present: Deep Learning is Everywhere

Medical Imaging



Levy et al, 2016

Galaxy Classification



Dieleman et al, 2014

From left to right: public domain by NASA, usage permitted by ESA/Hubble, public domain by NASA, and public domain.

Whale recognition



Kaggle Challenge

This image by Christin Khan is in the public domain and originally came from the U.S. NOAA.

2012 to Present: Deep Learning is Everywhere

Image Captioning

Vinyals et al, 2015
Karpathy And Fei-Fei, 2015



A white teddy bear sitting in the grass



A man in a baseball uniform throwing a ball



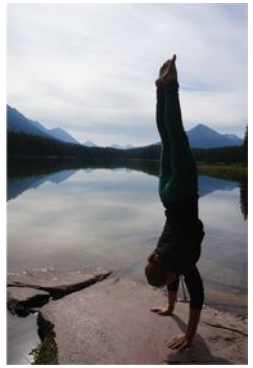
A woman is holding a cat in her hand



A man riding a wave on top of a surfboard

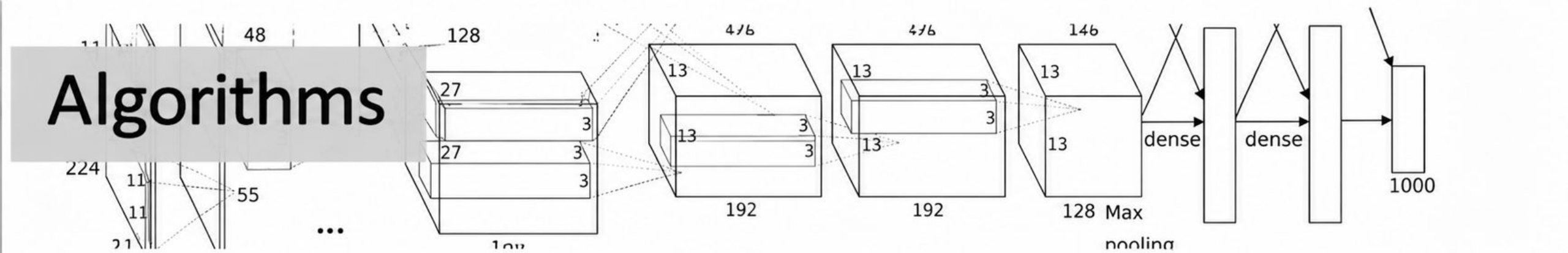


A cat sitting on a suitcase on the floor



A woman standing on a beach holding a surfboard

Algorithms



Data



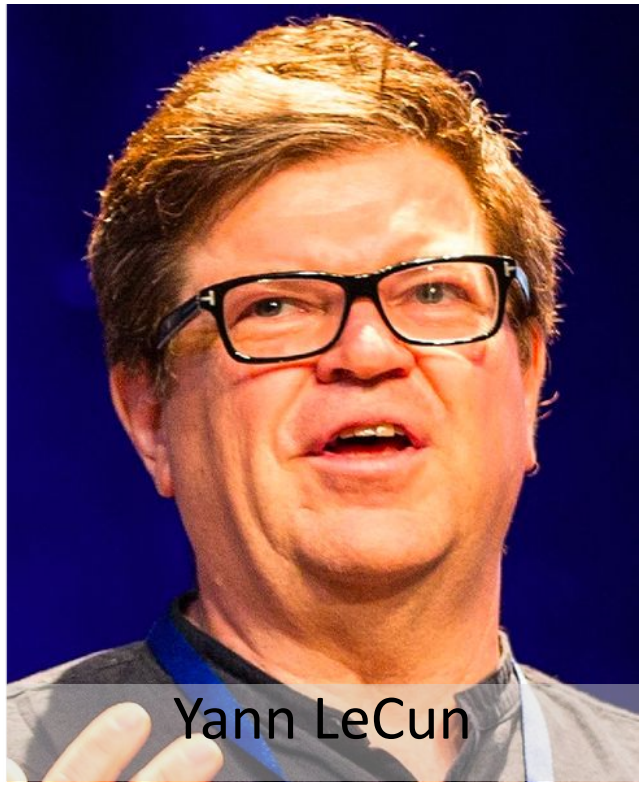
Computation

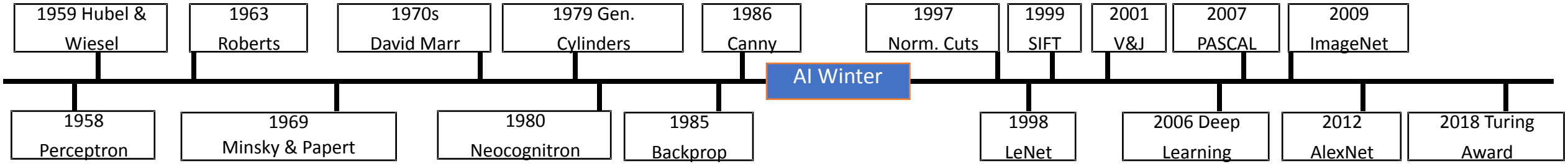


2018 Turing Award


Yoshua Bengio

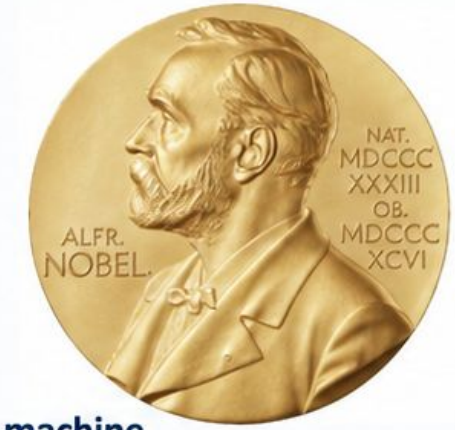

Geoffrey Hinton


Yann LeCun



2024 Nobel Prize in Physics

"for foundational discoveries and invention that enable machine learning with artificial neural networks"

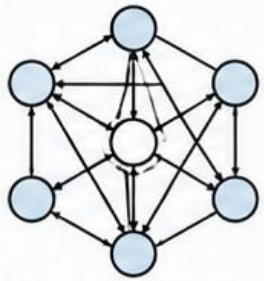


THE NOBEL PRIZE IN PHYSICS 2024



John J. Hopfield
Princeton University, USA

Hopfield network (1982)

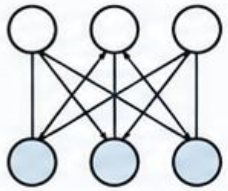


- Associative memory
- Energy-based model
- Foundations for modern neural networks



Geoffrey Hinton
University of Toronto, Canada

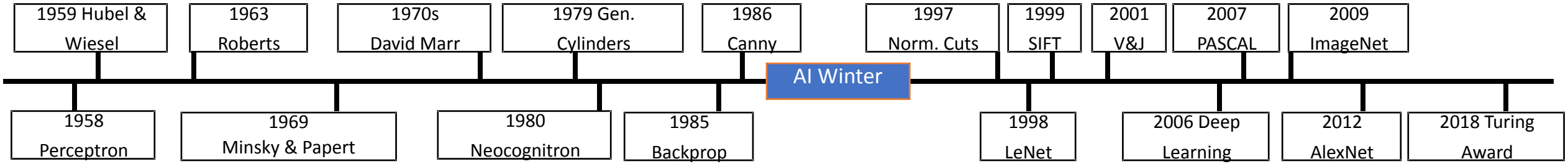
Boltzmann machine (1985)



- Probabilistic generative model
- Learning with uncertainty
- Key ideas connecting statistical physics and machine learning

"Their discoveries have been of the utmost importance for the development of machine learning with artificial neural networks."

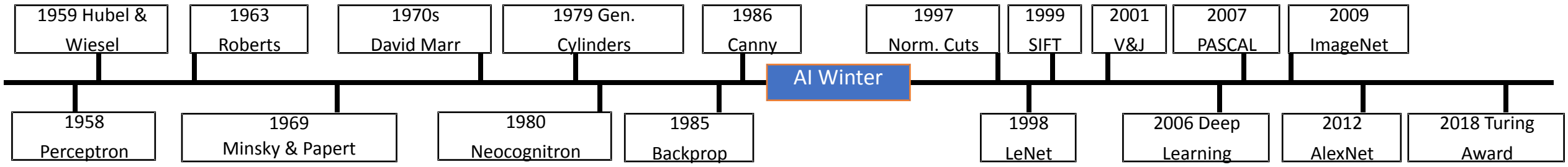
— The Royal Swedish Academy of Sciences




Autonomous Driving
Computer Vision Technology
Can Better Our Lives
地铁
Subway
Subway
地铁
Live Translation
- Hello
你好
- How can I help you?
我能帮你什么?
A golden retriever
in the style of
Van Gogh
AI Assistance
Tumor detected
Segmentation
Quantification
Report ready
Similar products
How can
I help you today?
Snow Leopard 0.98

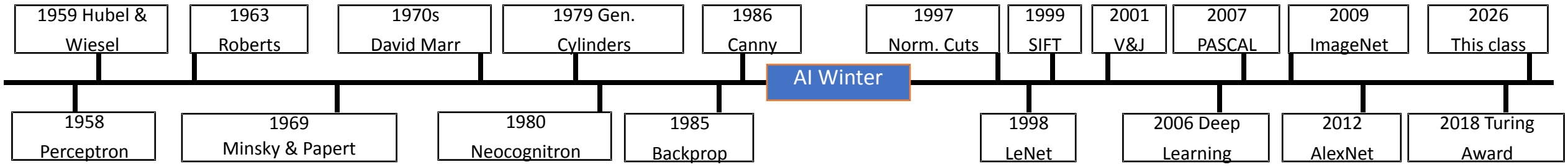
Today's Agenda

- A brief history of computer vision and deep learning
- Course logistics



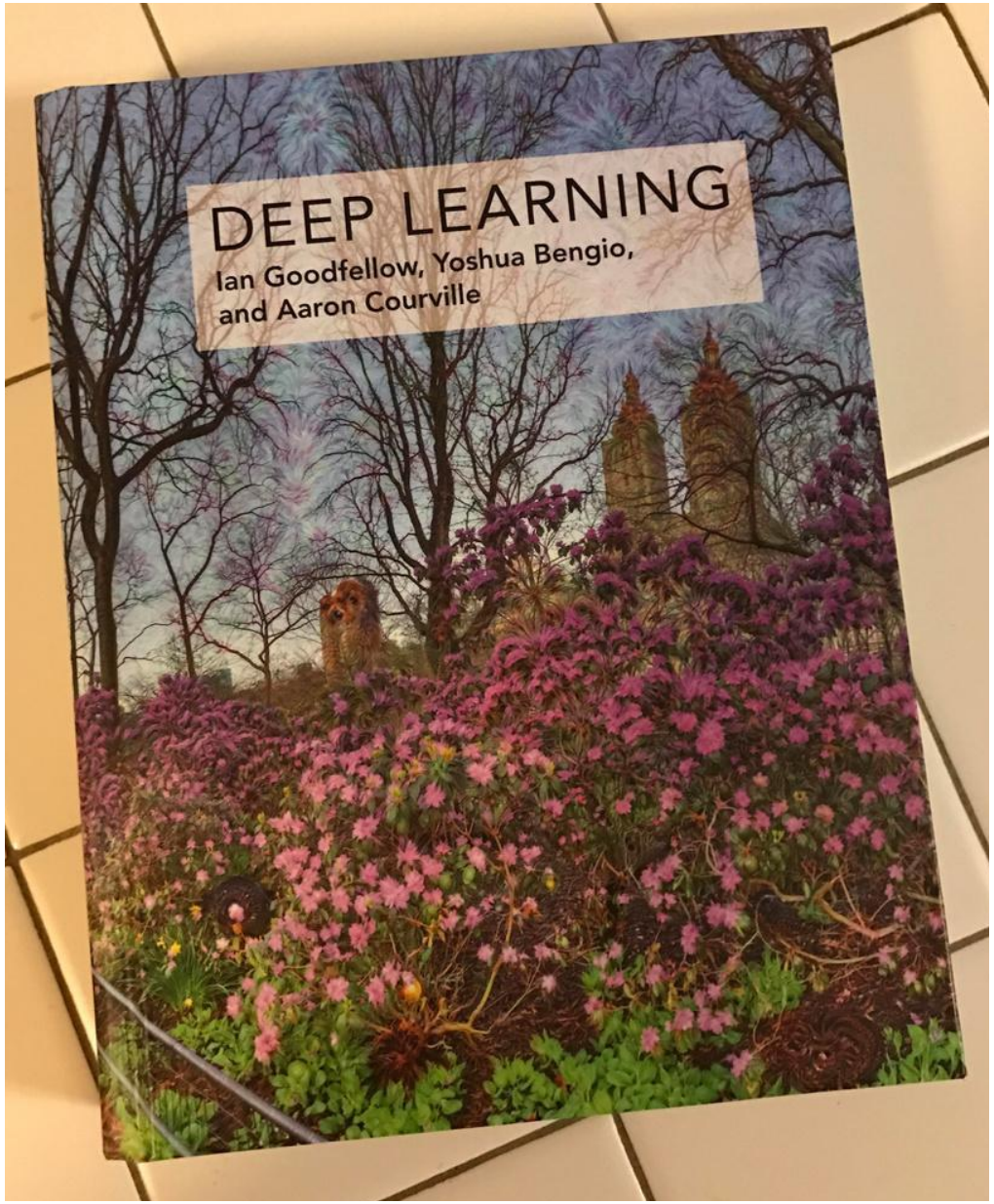
Today's Agenda

- A brief history of computer vision and deep learning
- Course logistics



Optional Textbook

- *Deep Learning* by Goodfellow,Bengio, and Courville
  Free online



Course Content

https://github.com/naziherrahel/Deep-Learning-for-Computer-Vision

First homework assignment

- Look for a research paper that had an impact on the computer vision field
- Create a ppt about the chosen paper

Next Lecture: Image Classification